MILITARY SPECIFICATION
CONVERTER-DISPLAY CV-3252/A

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, manufacture, and acceptance requirements for CV-3252/A Converter-Display, herein referred to as the converter-display, and converter-display mounting tray.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-K-3926	Knobs, Control (for use with Electronic, Communication and Allied Equipment)
MIL-W-5088	Wiring, Aircraft, Selection and Installation of
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of
MIL-M-18012	Marking for Aircrew Station Displays, Design and Configuration of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-M-38510	Microcircuit, General Specification for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect,(Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-81774	Control Panel, Aircraft, General Requirements for
MS-27651-03-C	Meter, Time Totalizing, Miniature, Digital, 115V, 400 Hz
MS90451-7152	Lamp, Incandescent, T-1 Bulb, Short, Integral Lighting
MS90452-7153	Lamp, Incandescent, T-3/4 Bulb, Integral Lighting

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STANDARDS

FEDERAL

FED-STD-595	Federal Standard Colors

MILITARY

MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-783	Legends for use in Aircrew Stations and on Airborne Equipment
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Method
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1390	Level of Repair
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1474	Noise Limits for Army Material
MIL-STD-2068	Reliability Development Test for Avionic Equipment

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MIL-STD-2074	Failure Classification for Reliability Testing, General Requirements for
MIL-STD-2076	VAST System/Avionics System Compatibility, General Requirements for
MIL-STD-2084	Maintainability of Avionic Equipment and Systems, General Requirements for

2.1.2 Publication. The following publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATION

NAVAL AIR ENGINEERING CENTER

NAEC-ACEL-533	Anthropometry for Naval Aviators - 1964

(Copies of specifications, standards, and publications required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DRAWINGS

2845951	Converter-Display Outline/Installation Drawing, Mounting Tray
2845952	Converter-Display Outline/Installation Drawing

(Application for copies should be addressed to the IBM Corporation, LAMPS MK III Program Control Office, Federal System Division, Owego, New York 13827.)

STANDARD

ELECTRONIC INDUSTRIES ASSOCIATION

RS-343	Hi-Resolution Monochrome Video Format

(Application for copies should be addressed to Electronic Industries Association, 2001 I Street N.W., Washington, D.C. 20006.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for the associated

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detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Parts, materials, and processes. Selection of materials and processes and approval of nonstandard materials and processes shall be in accordance with MIL-E-5400. Selection and approval of parts shall be in accordance with MIL-E-5400 except as provided in 3.2.1 and 3.2.2.

3.2.1 Parts selection. Parts selection shall be in compliance with MIL-E-5400 and the following:

3.2.1.1 Microelectronics. Microelectronic devices and microelectronic modules shall be hermetically sealed. Interchangeability shall be in accordance with MIL-STD-965, except that standard microelectronic parts as specified in 3.2.2 shall not require a demonstration of interchangeability. Microcircuits shall conform to MIL-M-38510 with the exception that organic or polymeric materials may be allowed inside hybrid packages when approved as a nonstandard part. JAN parts, Class B shall be used, if available. Screening of microelectronics shall be in accordance with MIL-STD-883, Methods 5004 and 5005, Class B.

3.2.1.2 Semiconductors. In carrying out the parts selection procedures for semiconductors, preference shall be given to JANTX V devices. Where JANTX V devices are not available, the following order shall take precedence:

a. JANTX.

b. JAN screened to TX level.

c. Nonstandard screened to TX V level.

Semiconductors shall be hermetically sealed. Plastic encapsulated devices shall not be used.

3.2.1.3 Other part types. In carrying out the parts selection procedures, preference shall be given to Established-Reliability Military Specification parts.

3.2.2 Part approval. The contractor shall obtain from the procuring activity approval of all nonstandard parts used in the equipment. Standard microelectronic parts shall conform to the following requirements:

a. At least two qualified sources shall be listed on the current Qualified Parts List (QPL).

b. The part shall be procured from one of the qualified sources.

c. Quality level "B" (or better) of MIL-M-38510 shall be required.

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3.2.2.1 Approval procedures. The approval procedures shall be in accordance with MIL-E-5400 and MIL-STD-965, as applicable. In addition to the requirements of MIL-E-5400, approval of nonstandard parts will be based on the following:

a. Suitability for the application.

b. Conformance to this specification and to MIL-E-5400.

c. Interchangeability (for microelectronic devices and modules, interchangeability demonstration in accordance with MIL-STD-965 shall be provided for each application).

d. Screening (burn-in) method.

Nonstandard parts shall be approved in accordance with MIL-STD-965. Test data required shall be limited to proving the part acceptable for its application. It may consist of existing data, subject to determination by the procuring activity of its adequacy and applicability. Specifications and standards for MIL-STD-143 Group IV parts and microelectronic parts shall accompany the request for approval.

3.3 Design and construction. The converter-display shall conform to MIL-E-5400 for design, construction, and workmanship, except as noted in Appendix II, or otherwise specified herein. The electronic portions of the equipment shall be functionally modularized in accordance with MIL-STD-2084.

3.3.1 Total weight. The total weight of the converter-display and mounting tray shall be not greater than 38.2 pounds.

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a contractor who has previously produced the converter-display, the program previously used may be continued unless otherwise indicated in the contractor or order (see 6.2.1).

3.3.2.1 Operational stability. The converter-display shall operate with specified performance, continuously or intermittently, for not less than 2000 hours or one mean-time-between-failure (MTBF) whichever is greater, without the necessity for readjust-ment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating service life. The converter-display shall have an operating service life of 10,000 hours when operated in any operational mode or combinations of modes and under the environmental conditions specified herein, with no preventive maintenance and no scheduled parts replacement except as approved by the procuring activity.

3.3.2.3 Storage life. The converter-display shall be capable of withstanding an 18-month storage period under environmental conditions as specified herein without requiring any maintenance prior to operation.

3.3.2.4 Reliability in MTBF. The converter-display, including any Built-In-Test provisions, shall have a specified MTBF (as specified in MIL-STD-781) of 2600 hours when tested and accepted as specified in Section 4 of this specification. A failure shall be specified as any operation outside the performance limits specified herein, including any unscheduled adjustments or calibration (except normal operating adjustment of accessible controls).

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3.3.2.5 Time totalizing meter. The WRA, converter-display panel, shall contain time totalizing meters in accordance with MIL-M-7793 and MS-27651-03-C.

3.3.3 Cabling and connections.

3.3.3.1 Cables and connectors. The converter-display shall provide for the use of cables and connectors in accordance with MIL-E-5400. The connectors shall be in accordance with MIL-C-38999, Series I, Class T, Finish B, except J2 and J3 shall be rack and panel connectors. Use of connectors not covered by MIL-C-38999 shall require prior approval by the procuring activity. Connectors shall be located on the front surface of WRAs except for console- and rack-mounted panels.

3.3.3.2 Interconnection cabling. The converter-display shall be capable of specified operation using external wiring in accordance with MIL-W-5088. The external wiring shall be unshielded, except as required herein (see 3.4.2) and except that a minimum number of the individual wires or cables may be shielded when demonstrated as necessary to meet interference control requirements. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the converter-display.

3.3.4 Control panels. All rack- or console-mounted control panels shall conform to MIL-C-6781 and MIL-C-81774.

3.3.5 Interchangeability. The converter-display shall meet the interchangeability requirements of MIL-E-5400.

3.3.6 Electromagnetic interference.

3.3.6.1 Electromagnetic interference (EMI) control. The converter-display shall be designed in accordance with MIL-STD-461, except as modified by the following:

a. The radiated susceptibility levels of Requirement RS03 shall be as follows:

   (1) 14 kHz to 200 Hz - 40V/M.

   (2) 200 MHz to 10 GHz - 20V/M.

b. The permissible power line conducted narrowband emission levels of CE01 shall be tailored to the nominal operating current level at the primary power operating frequency. The EMI level permitted for the input current level shall be as shown on figure 1.

3.3.7 Provisions for maintainability. The maintainability design, built-in-test features, construction, packaging, provisions for test points, and other maintainability requirements shall be as specified in MIL-STD-2084 except as modified herein. Definitions of WRA, SRA and IPRA are in MIL-STD-2084.

3.3.7.1 Maintainability program. The contractor shall evaluate, design, and integrate the maintainability features of the converter-display by conducting a maintainability program in accordance with MIL-STD-470.

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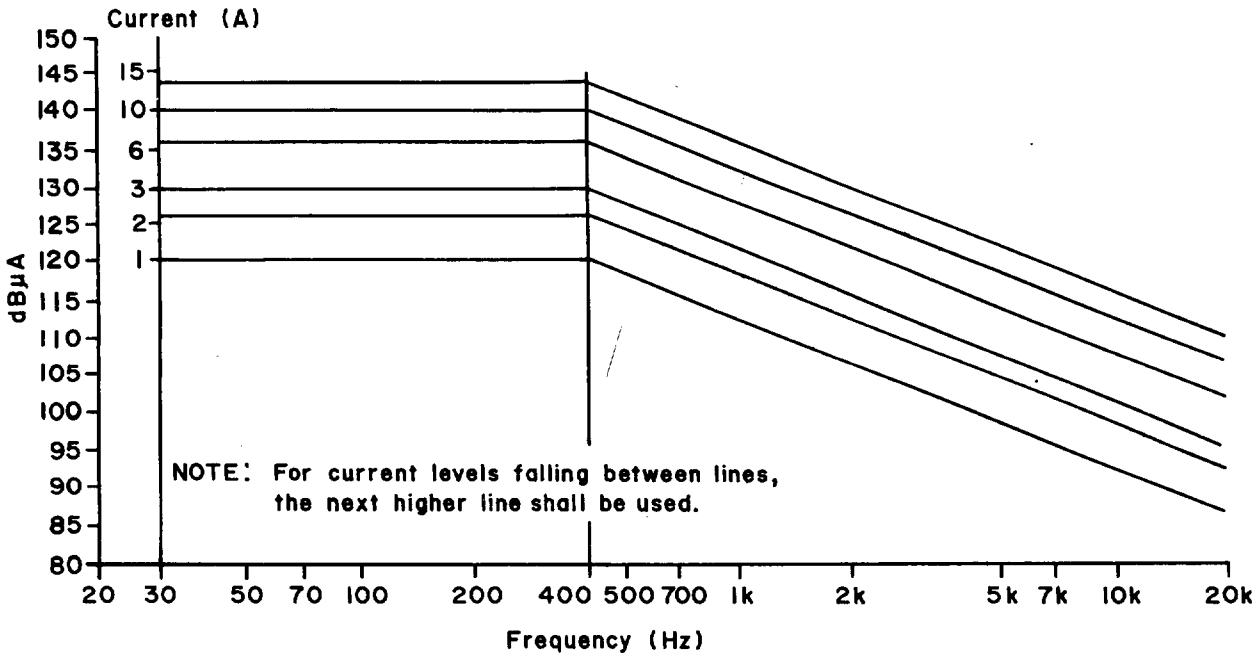


FIGURE 1. CE01 narrowband requirements.

3.3.7.2 Compatibility with ATE. The converter-display shall be compatible with Automatic Test Equipment (ATE) in accordance with the design requirements of MIL-STD-2076.

3.3.7.3 Mounting devices. Connectors, receptacles, fuses, circuit breakers or any other electronic component, shall not be a part of any provisions for mounting the converter-display.

3.3.7.4 Keying. Modules and connectors (with the exception of Radio Frequency (RF) type connectors) shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the converter-display. Internal RF connector cables shall be marked with legible reference designations. External RF connectors shall be color coded in accordance with MIL-W-5088.

3.3.7.5 Servicing. The converter-display shall meet the reliability and maintainability requirements herein with no scheduled servicing at any level of maintenance.

3.3.7.6 BIT implementation. In addition to MIL-STD-2084, the converter-display Built-In-Test (BIT) shall provide a system self-test, a monitor capability, and fault isolation to the WRA. The BIT circuits shall be protected against internally induced transients and MIL-STD-704 type line transients or interrupts to prevent false BIT indications.

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3.3.7.6.1 Self-test. The self-test shall be ready for external initiation upon application of power to the converter-display and shall require not more than one minute to execute, excluding warm-up time. Provisions shall be made for initiation of BIT via a test position of the power ON-OFF switch. As specified in MIL-STD-2084, the converter-display failure detection requirements modified to read 95 percent of failures to be detected rather than 98 percent, shall apply to all equipment failures except failures of the following functions:

a. BIT and ATE circuitry.

b. Non-effect protection circuitry.

c. Panel lights.

d. Elapsed time indicator.

Operator evaluation of the internally generated converter-display test pattern will be permitted as a means to detect faults. Fan failures shall be determined by operation of the overtemperature sensor. Additional failures shall be detectable using an off-line converter-multiplexer (CMUX) and signal data converter (SDC) assisted manual test as required to achieve 98 percent detection of failures, with exclusions as above. The BIT self-test features shall be such as to eliminate the need for any auxiliary test device to perform fault detection or isolation other than the CMUX or SDC as noted above.

3.3.7.6.2 Status monitor. In normal operating mode the monitor capability shall be initiated by periodic receipt of a differential discrete signal from the CMUX and shall operate while the converter-display is performing the specified mission function. As specified in MIL-STD-2084, the converter-display failure detection requirements modified to read 95 percent of failures to be detected rather than 98 percent, shall apply to all converter-display failures except failures of the following functions:

a. BIT and ATE circuitry.

b. Non-effect protection circuitry.

c. Panel lights.

d. Elapsed time indicator.

Operator evaluation of the operational display will be permitted as a means to detect faults. Fan failures shall be determined by operation of the overtemperature sensor.

3.3.7.6.3 Fault isolation. The converter-display BIT shall isolate the faulty WRA in accordance with MIL-STD-2084.

3.3.7.7 Fault indicators. The WRA, converter-display shall contain a fault indicator which shall hold the last test result. The fault indication shall be black for a "GO" condition and white for a "NO-GO" condition and shall not require manual resetting. The fault indicator shall be located on the front panel of the converter-display.

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3.3.7.8 Personnel. The converter-display shall be designed such that all maintenance actions can be performed by a technician with a comprehension ability level as follows:

a.	Civilian Education	Ninth grade education or equivalent.
b.	Technical Training	Navy, Class A and Class C school appropriate to the maintenance task.
c.	Experience	One year of technical experience in addition to technical training.

3.3.7.9 Mean-Time-To-Repair, organizational level. The Mean-Time-To-Repair (MTTR) shall include:

a. Preparation time (turn on, warm-up).

b. Fault detection.

c. Fault isolation to a WRA.

d. WRA replacement.

e. Checkout of repair.

MTTR shall be the summation of all maintenance downtimes listed above divided by the number of maintenance actions performed. For Organizational Maintenance, the MTTR shall be not greater than 15.0 minutes. The Maximum time required to accomplish any single maintenance action, M(maxct), at the Organizational Level shall be not greater than 30.0 minutes at the 95th percentile.

3.3.7.10 Intermediate level maintenance. Fault detection and isolation to the SRA at this level shall be accomplished without the removal of any converter-display covers or panels. All test points shall be available in a connector(s) located on the external surface of the equipment. An environmental protection cover shall be provided on the test connector. No WRA shall be dependent upon any other WRA for the purpose of testing, fault detection, and fault isolation.

3.3.7.10.1 SRA accessibility. Module arrangement shall be such that access to any module does not require the removal of adjacent modules or parts other than access panels except for Cathode Ray Tube (CRT) removal. No special tools shall be required to remove and replace SRAs, or repair an IPRA.

3.3.7.10.2 SRA access. Access panels and dust covers (for access to the SRA) shall be retained by quick disconnect captive fasteners, for example, hand-operated slide or DZUS type. The use of quick disconnect devices shall not prevent the converter-display from meeting the EMI requirements of this specification.

3.3.7.10.3 SRA repair. All SRAs shall be designed to facilitate repair. Part replacement on the modules shall be possible by technicians working in a service area such as intermediate maintenance activity. SRAs other than the harness terminations, High Voltage Power Supply (HVPS), and CRT assemblies shall not be encapsulated, potted or embedded. When conformal

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coating is used, it shall be a transparent type which is easy to remove and reapply without damage to the SRA. Disposition (repair vs throwaway) of faulty SRAs shall be established using level of repair analysis performed in accordance with
MIL-STD-1390.

3.3.7.10.4 Chassis-mounted components. No component shall be mounted to the converter-display or WRA chassis, except when that component provides a direct mechanical, electrical or visual external interface.

3.3.7.10.5 Mean access and repair time - intermediate level. The mean access and repair time shall include the following:

a. SRA/chassis-mounted component access.

b. SRA/chassis-mounted component remove/replace.

c. WRA close up.

Mean access and repair time shall be the summation of all maintenance downtimes listed above divided by the number of maintenance tasks performed. For intermediate maintenance, the mean access and repair time shall be not greater than 15 minutes. The maximum time required to accomplish any single maintenance action, M(maxct), at the Intermediate Level shall be not greater than 30 minutes at the 95th percentile.

3.3.7.10.6 Shop fixed interface (SFI) ratio. The ratio of the number of SRAs which do not require adjustment or trimming at installation in the WRA to the total number of SRAs shall be not less than 0.50.

3.3.7.11 Handles and grasp areas. The WRAs shall be designed (in accordance with MIL-STD-1472) to have handles or other suitable means for grasping, handling, and carrying by gloved or mittened hand except that the size of the handles shall be as shown on figure 3 rather than as specified in MIL-STD-1472.

3.3.8 Nomenclature, nameplates, and identification marking. Nomenclature and serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions:

a.	Temperature	Room ambient ($25^{o}C \pm 10^{o}C$)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power Voltage	115±1.0 Vac, 400 Hz, 3 Phase Panel Lighting Voltage 5±0.1 Vac, 400 Hz

3.3.10 Service conditions. The converter-display shall operate normally under any of the environmental service conditions, or combination of these

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conditions, as specified in MIL-E-5400 for Class 1B equipment, except as modified herein.

3.3.10.1 Temperature.

3.3.10.1.1 High and low temperature (operating). The converter-display shall operate during and after being subjected to temperature exposure in accordance with 4.4.3.1.1. Low temperature extreme shall be -40°C. High temperature extreme shall be +71°C for continuous operation and +95°C for intermittent operation.

3.3.10.1.2 High and low temperature (storage). The converter-display shall operate after being subjected to temperature exposure (non-operating) in accordance with 4.4.3.1.2. Low temperature extreme shall be -62°C, and high temperature extreme shall be +95°C.

3.3.10.1.3 Temperature shock. The converter-display shall operate and shall not exhibit physical damage after being exposed to temperature shock test in accordance with 4.4.3.8.

3.3.10.2 Low pressure (altitude). The converter-display shall operate and shall not exhibit physical damage during and after being subjected to altitude exposures in accordance with 4.4.3.2. Altitude extreme shall be 40,000 feet.

3.3.10.3 Explosion proof. The converter-display shall not cause an explosion when operated in an explosive atmosphere from sea level up to 15,000 feet altitude in accordance with 4.4.3.7.

3.3.10.4 Salt fog. The converter-display shall operate and shall not exhibit physical damage after being subjected to salt fog exposure in accordance with 4.4.3.6.

3.3.10.5 Vibration. The converter-display, excluding fasteners, shall operate when subjected to the vibration requirements of Curve IIIb of MIL-E-5400, except that the lower frequency shall extend to 4 Hz, and between 4 Hz and 7 Hz, the acceleration shall be 0.5g. The converter-display shall be designed such that no resonant frequency exists below 50 Hz. Resonance is specified as an amplification of 2 or greater between the input vibration level and the response at any point on or within the converter-display. Vibration isolators shall not be used. A notch will be allowed from 5g to 4g in the frequency band of 200±30 Hz.

3.3.10.6 Shock.

3.3.10.6.1 Functional shock. The converter-display shall operate and shall not exhibit physical damage during and following functional shock tests in accordance with 4.4.3.4.

3.3.10.6.2 Crash hazard shock. The converter-display shall not exhibit physical damage following crash hazard shock test in accordance with 4.4.3.4.

3.3.10.6.3 Bench handling. The converter-display shall meet specified performance and accuracy requirements after being subjected to bench handling test in accordance with 4.4.3.4.

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3.3.10.7 Humidity. The converter-display shall meet the operational requirements of this specification after being subjected to a humidity test in accordance with 4.4.3.5. The converter-display shall not be damaged as a result of the humidity test.

3.3.11 Warm-up time. The time required for the converter-display to warm-up prior to operation shall be not greater than 2 minutes under standard conditions and 5 minutes at extreme service conditions.

3.3.12 Input electrical power.

3.3.12.1 Operating power. The converter-display shall conform to MIL-STD-704 and shall give specified performance when energized from ac power (three phase) 115 V/400 Hz, Category B, 275 VA power sources having characteristics and limits as specified in MIL-STD-704.

3.3.12.2 Lighting power. Input power for lighting shall require not more than the 5 VA at 5 volts ac.

3.3.13 Cooling air. Cooling shall be limited to the use of ambient air identical to the service conditions of 3.3.10. All cooling provisions shall be integral to the converter-display with entrance at the sides and top and exit at the rear. The converter-display shall be designed using cold plates or heat exchangers so that none of the cooling air will come in contact with internal parts, circuitry or connectors.

3.3.13.1 Overtemperature sensor. An overtemperature sensor shall be provided to cause an output within $5^{o}C$ of maximum permissible internal operating temperature of the converter-display. Operation of the overtemperature sensor shall not result in automatic converter-display turnoff. Upon actuation, the hysteresis of the sensor shall be such that the device shall reset within 10 minutes after restoration of environmental service conditions (see 3.3.10) within the converter-display. These switched contacts shall be routed to two pins on the J2 connector.

3.3.14 Finish. Exterior surfaces of the converter-display (see 3.5.1) and converter-display mounting tray (see 3.5.2) shall be finished in lusterless black, Color Number 37038, in accordance with FED-STD-595.

3.3.15 Human factors engineering. Unless modified by Appendix B or otherwise specified in the contract (see 6.2.1), aircrew station displays, controls, panels, and avionic subsystems shall conform to the design criteria of MIL-STD-1472 to assure adequate operability and maintainability. Designs shall be compatible with the 3rd - 98th percentile operator as specified in NAEC-ACEL-533, and with the 5th - 95th percentile maintenance man as specified in MIL-STD-1472.

3.3.15.1 Aircrew station displays, controls, and panels. The design and formatting of the aircrew station displays, controls, and panels shall be in accordance with the design criteria and requirements of MIL-STD-1472 and MIL-P-7788.

3.3.15.2 Labeling. The labeling of displays and controls on panels shall be in accordance with the requirements of MIL-M-18012 and MIL-STD-783, except as modified by 3.3.15. The Futura demibold letter design with three heights (0.125, 0.156, and 0.188 inch) shall be selected from MIL-M-18012. A constant character-height to stroke-width ratio of 6.5 ± 0.5 shall be used.

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3.3.16 Acoustical noise. The acoustical noise output of each converter-display shall be not greater than 73 dB(A) as specified in MIL-STD-1474, when measured at a distance of 1 meter perpendicular to the center of any unit surface when the converter-display is located in a semi-anechoic chamber.

3.3.17 Safety. Converter-display design safety characteristics shall be in accordance with MIL-E-5400. X-radiation producing equipment must be designed to limit the personnel dosage rate at any readily accessible point, measured two inches from the outer surface, to 0.5 millirem per hour with external shielding removed. This applies to the areas normally inhabited by personnel during installation, calibration, maintenance and operation of the converter-display.

3.4 Performance. Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the external conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation. The converter-display shall be a self-contained unit using a cathode ray tube to produce the display images, in response to input deflection and video signals from the CMUX or composite video from the SDC. The converter-display shall meet the visual and electrical requirements of 3.5.

3.4.1.1 Display modes. The converter-display shall provide a symbol display mode or raster display mode or, through time-multiplexed input signals, operate in both modes. The symbol mode display will depict the tactical and navigation situation and the raster mode will display sensor equipment information.

3.4.1.2 Operating modes. The converter-display shall provide two operational modes, normal mode and backup radar mode. In normal mode, the converter-display shall be controlled by the CMUX via all interface signals shown on figure 2. In backup radar mode all CMUX/converter-display interface signals shall be deselected and the converter-display will process the composite video signal. Operating mode selection shall be provided by the backup radar mode select discrete from the SDC.

3.4.2 Contents. The converter-display shall contain all power supplies, components and circuitry necessary to perform the specified functions in a single WRA.

3.4.3 Controls. The controls on the converter-display shall permit manual adjustment of visual and electrical parameters. Control knobs shall be selected in accordance with MIL-K-3926.

3.4.4 Interface diagram. An interface diagram showing all converter-display interfaces with associated LAMPS equipment is shown on figure 2.

3.5 Detail requirements.

3.5.1 Converter-display. The converter-display shall meet the requirements of 3.5.1.1 through 3.5.1.15.

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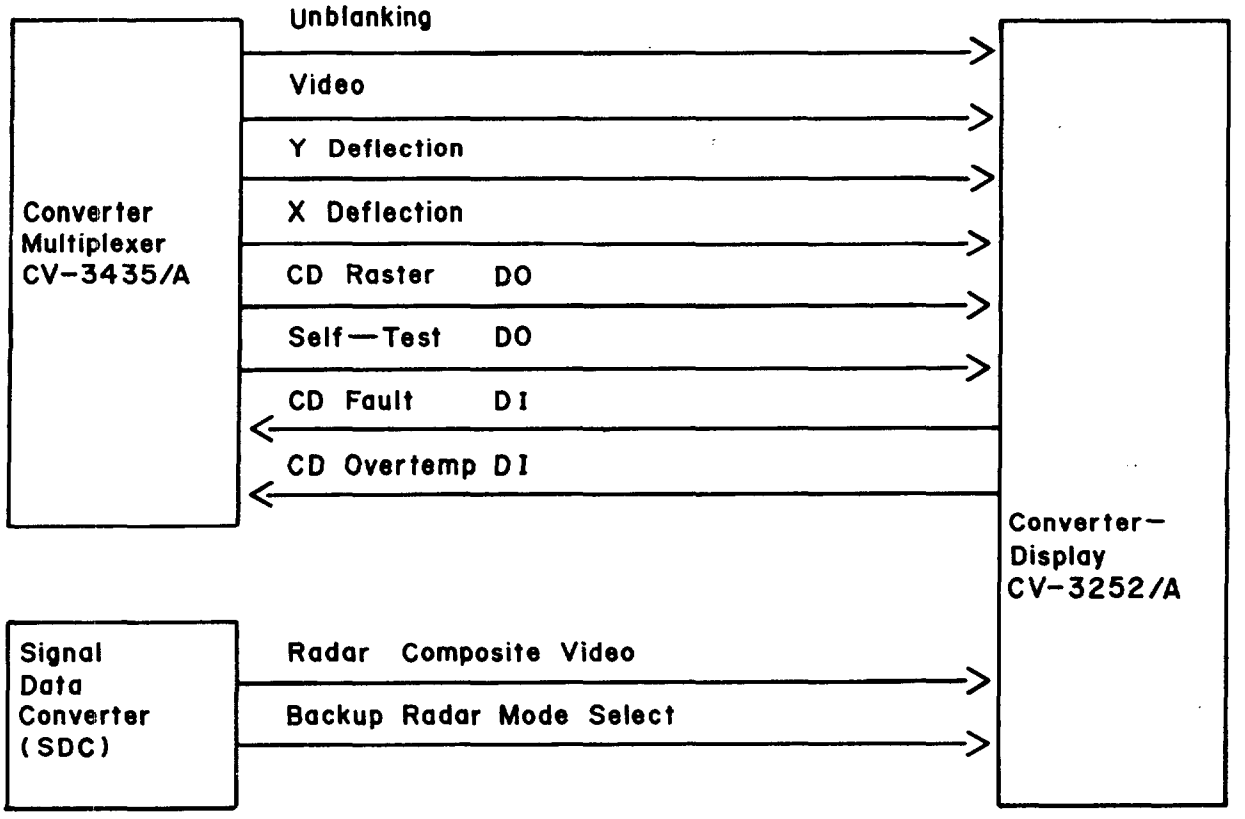


FIGURE 2. Converter-display interface diagram.

3.5.1.1 Function. The converter-display shall provide the operator with presentations of tactical, navigation and sensor data. The tactical and navigation situations will be depicted in a symbol mode. The symbol mode shall also be capable of presenting information by alphanumerics in tabular or text form. The raster mode shall be capable of presenting the sensor equipment information.

3.5.1.2 Form factor. The converter-display form factor shall be in accordance with Drawing 2845952 and that shown on figure 3. The size of the converter-display excluding connectors, integrally illuminated panel, controls and other functional hardware shall be not greater than the following dimensions:

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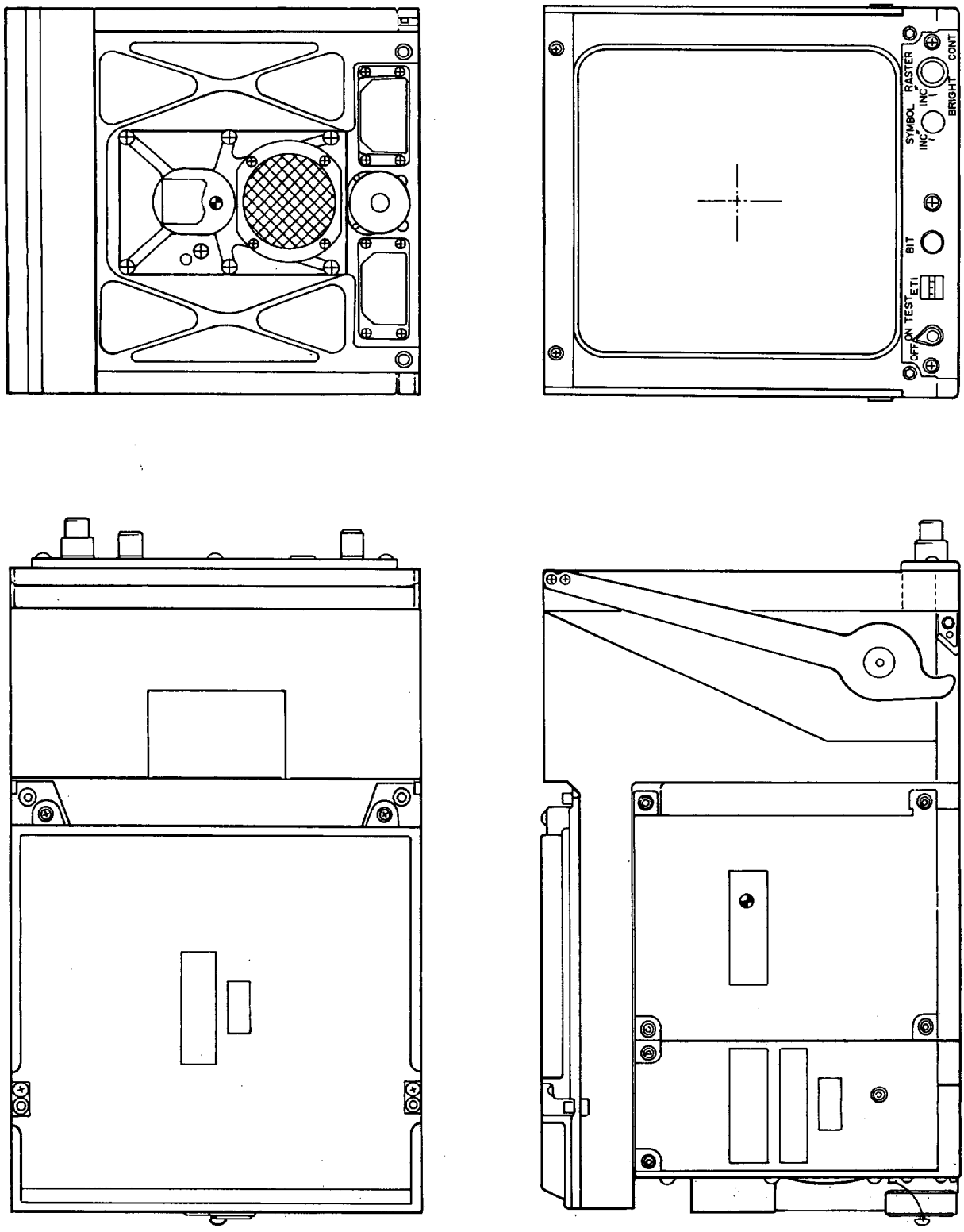


FIGURE 3. Outline/installation drawing, converter-display.

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a. Width 9.1 inches

b. Height 9.06 inches

c. Depth 14.4 inches (front surface to back surface of box)

3.5.1.3 Weight. The total weight of the converter-display excluding cables and mounting tray shall be not greater than 36.4 pounds.

3.5.1.4 Controls and indicators. Controls, connectors, and other functional hardware shall not protrude more than 1.0 inch beyond the front surface nor more than 1.25 inches beyond the rear surface except the raster intensity control may not protrude more than 1.6 inches beyond the converter-display front surface. The converter-display shall be capable of installation and removal by sliding in and out of its mounting tray. The converter-display shall engage its system connector when installed. An integral carrying handle on the converter-display shall jack the converter-display into and out of the mounting tray and connector. The converter-display shall be designed to be mounted without vibration isolators (hard mounted). The converter-display without mounting tray is shown on figure 4.

3.5.1.4.1 Carrying handle. The carrying handle shall not protrude beyond the front panel of the converter-display when the converter-display is in the installed position. The handle shall be designed so that it can be locked in place.

3.5.1.5 Shop repairable assemblies (SRAs). The converter-display shall contain the following SRAs:

a. Low voltage power supply (see 3.5.1.5.1).

b. Cathode ray tube (CRT) assembly (see 3.5.1.5.2).

c. HVPS (see 3.5.1.5.3).

d. Edge lighted panel (see 3.5.1.5.4).

e. Circuit card assembly (see 3.5.1.5.5).

f. RFI filter.

g. Wired structure.

3.5.1.5.1 Low voltage power supply (LVPS). The LVPS shall consist of power transistors, power resistors, a transformer, rectifiers, and filters necessary to generate all low voltages required.

3.5.1.5.2 Cathode ray tube assembly. The cathode ray tube assembly shall consist of tube, yoke and electromagnetic shield. The assembly shall be aligned and potted to preserve alignment and to reduce shock and vibration to the CRT.

3.5.1.5.3 HVPS. The HVPS shall furnish the anode, focus, and grid-2 voltages required by the CRT assembly and shall interface via connector assemblies.

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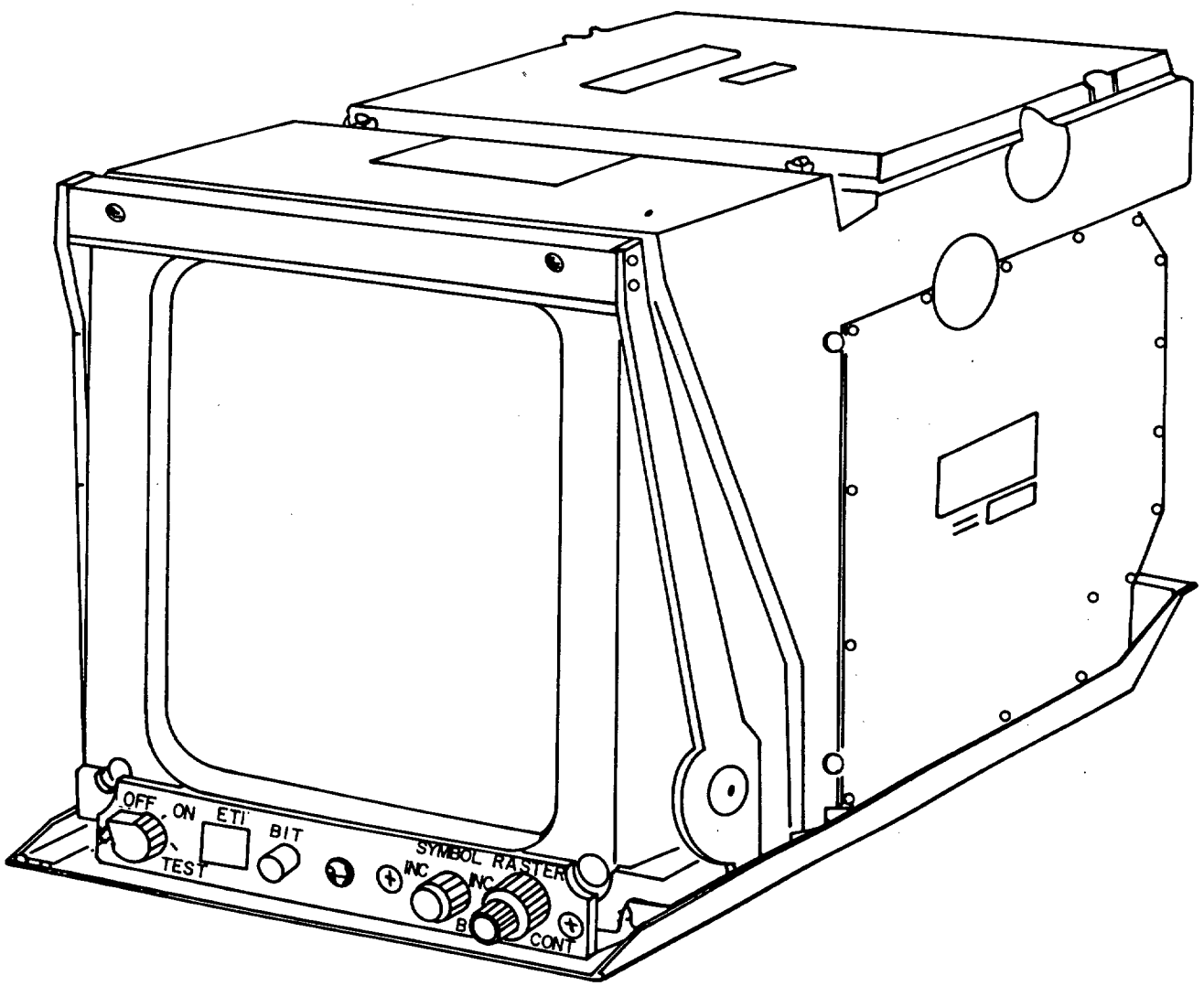


FIGURE 4. Converter-display.

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3.5.1.5.4 Lighting panel assembly. The front panel assembly shall consist of a plastic lighting panel.

3.5.1.5.5 Printed circuit board assembly. The printed circuit board assembly shall contain the deflection, video, I/O, self-test, and composite video processing functions.

3.5.1.5.6 Implosion protection. The converter-display shall be designed, fabricated, shielded and operated so that personnel shall be protected from implosion of the CRT. Operating the converter-display shall not require any special procedures. Handling, storage, and maintenance procedures must be acceptable to the procuring activity.

3.5.1.5.7 Arc suppressors. Arc suppressors shall be provided to protect low voltage circuitry from the high voltage.

3.5.1.6 Controls. The front panel layout shall be as shown on figure 5 and shall contain the following controls:

a. OFF-ON-TEST control. The OFF-ON-TEST control shall be a vertically-mounted three position rotary switch which shall apply or remove the 400 Hz, 3-phase, 115 V power source. In the "TEST" position, the test channel shall initiate the manual and periodic BIT and generate the test pattern presentation specified in 3.5.1.12.4.1

b. SYM INTENSITY control. The SYM INTENSITY control shall provide gain control for the symbol video, with at least 300^{0} of rotation while in the symbol mode of operation.

c. RASTER BRIGHT control. The RASTER BRIGHT control, located on the outer ring concentric knob, shall provide independent brightness control for the raster mode of operation.

d. RASTER CONTRAST control. The RASTER CONTRAST control shall be the center knob of a concentric control which shall control video gain while in the raster mode of operation.

3.5.1.6.1 RASTER BRIGHT control dynamic range. In raster mode a shift in cut-off voltage shall be provided to expand the dynamic range of the CRT cathode to g_1 voltage from full brightness to approximately 10 Vdc into the black region beyond CRT brightness cut-off. The shift in cut-off voltage shall allow the bright control to eliminate low level video or background noise (referred to as GREEN FOG) from being presented on the display at the minimum raster bright control setting. When adjusted together with the RASTER CONTRAST control the RASTER BRIGHT control shall provide successive elimination of up to 8 displayed video shades of gray to the black level.

3.5.1.7 Electrical connections. Connections to all external circuits shall be as specified in table I.

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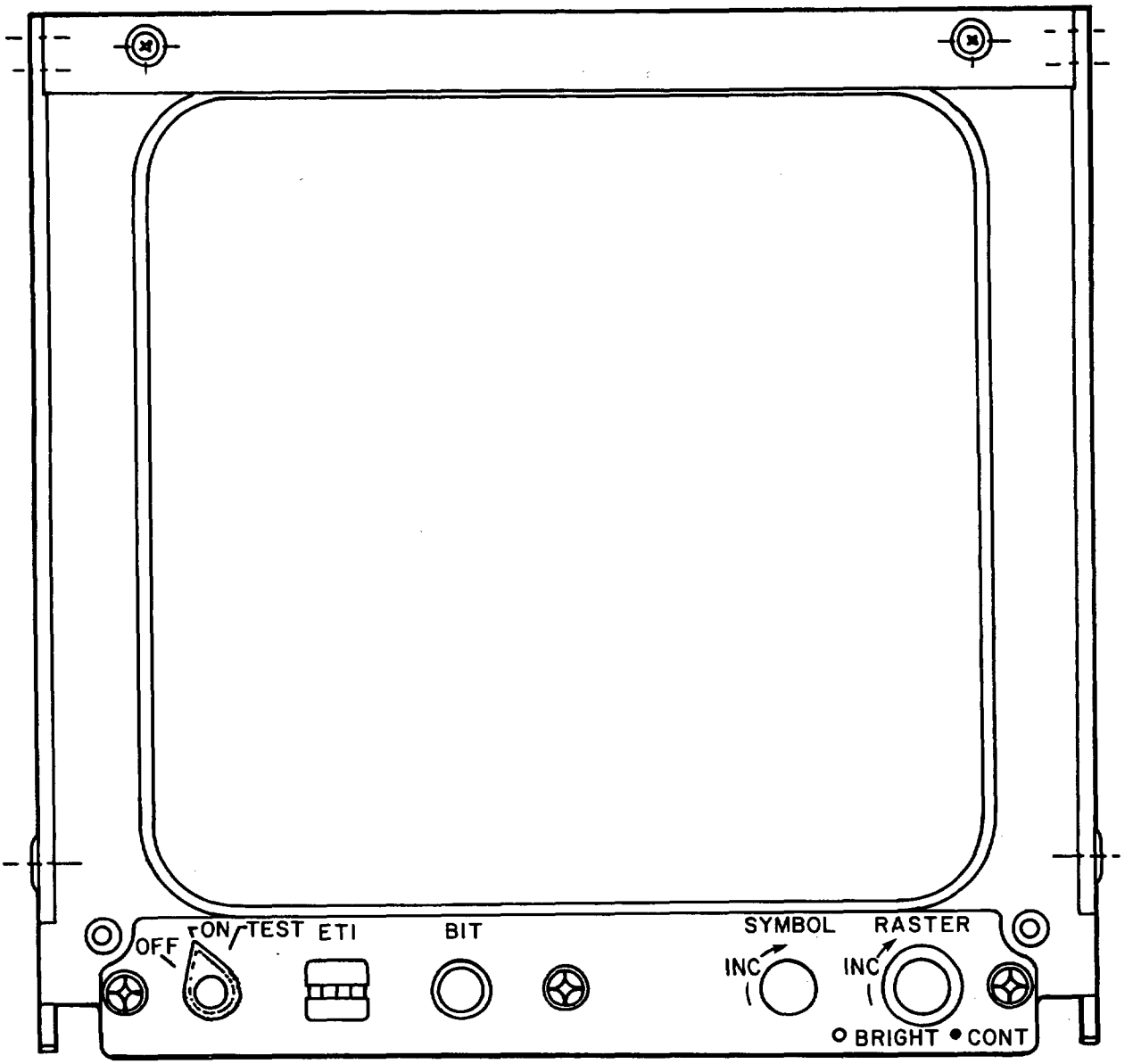


FIGURE 5. Converter-display (front view).

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TABLE I. Electrical connections.

Reference Designation	Receptacle Type	Function
J1	MS27656T19B35A	Test connector
J2	DPXAME-32W4-33S-FO	Signal
J3	DPXAME-32W4-34P-FO	Power

External interface signals, power connections, and test connector signals are specified in Appendix A, tables VI, VII, and VIII.

3.5.1.8 Display area. The converter-display shall have a usable display area of not less than 6.50 inches by 6.50 inches. This complete usable display area shall be viewable at any angle from a normal to the front surface of the high contrast filter as follows:

a. 30° minimum on both sides.

b. 15° minimum toward the top edge.

c. 10° minimum toward the bottom edge.

3.5.1.9 Front panel design. The light of the plastic panel shall conform to MIL-P-7788, Class 1-W, (IPL white in accordance with MIL-C-25050) Type V with an externally controlled lamp voltage of 0 to 5±0.10 Vac. The lamps used for illuminated panels shall be MS90451-7152 or MS90452-7153 in accordance with MIL-P-7788 and shall exclude any color correcting filters but shall include circuit board construction with the lamps soldered to the lands. The circuit board shall be recessed into the back of the plastic plate to prevent light leaks. The configuration of all panels must have prior approval of the procuring activity. Panel marking and control selection shall be in accordance with MIL-C-81774 except the concentric Raster Bright/Contrast Control. Touch temperature shall be in accordance with MIL-STD-454, Requirement 1.

3.5.1.10 Internal adjustment. The following continuously variable controls shall be provided as manual adjustments internal to the converter-display. Access to all internal adjustments will be by removal of the top or side covers.

a. Horizontal Gain (±3 dB about nominal gain).

b. Horizontal Centering (±0.35 inch from display center).

c. Vertical Gain (±3 dB about nominal gain).

d. Vertical Centering (±0.35 inch from display center).

e. Focus Adjust (.013 inch nominal spot size).

f. G-2 Adjust (400-800 Vdc).

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3.5.1.11 External input signals. External input signals consist of those signals specified in 3.5.1.11.1 through 3.5.1.11.1.4.

3.5.1.11.1 CMUX horizontal and vertical deflection signals. The horizontal and vertical deflection signals are used to move the electron beam across the face of the CRT along the horizontal and vertical axes respectively. The input deflection signals shall control electron beam in both the raster and symbol modes.

3.5.1.11.1.1 Signal amplitude. A +7 Vdc deflection signal on both channels shall deflect the electron beam to the upper right corner of the display, 0V on both channels shall center the beam on the display and -7 Vdc on both channels shall reflect the beam to the lower left corner of the display.

3.5.1.11.1.2 Signal to noise ratio. The signal to noise ratio of the input deflection signal will be 57 dB minimum with respect to +5V peak at 93 ohms.

3.5.1.11.1.3 Waveform. Typical input deflection signal waveforms for both symbol and raster modes are shown on figures 6 and 7, respectively.

3.5.1.11.1.4 Input impedance. The horizontal and vertical deflection channel input impedance shall be 93±3 percent ohms.

3.5.1.11.2 CMUX video signal. The video input signal is used to generate the required video in both raster and symbol modes.

3.5.1.11.2.1 Signal amplitude. The full-off or black level will be 0±.05V. The full-on or white level shall be +4V±0.2V. The video blanking level will be 0±0.2V.

3.5.1.11.2.2 Bandwidth. The video signal shall exhibit a bandwidth of dc to 27 MHz at the 3 dB point.

3.5.1.11.2.3 Signal to noise ratio. The signal to noise ratio will be 38 dB minimum with respect to 4V peak at 93 ohms.

3.5.1.11.2.4 Waveform. Typical waveforms of the input video signal are shown on figures 6 and 7.

3.5.1.11.2.5 Input impedance. The video signal channel input impedance shall be 93±3 percent ohms.

3.5.1.11.3 CMUX unblanking signal. The unblanking input signal shall provide the on/off and/or bias control of the CRT video signals during both raster and symbol modes.

3.5.1.11.3.1 Signal characteristics. The unblank signal shall be implemented with a balanced differential line conforming to table II. The blank level shall be a logic "0" and the unblank level shall be a logic "1". Figure 8 shows the required differential receiver biasing.

3.5.1.11.3.2 Waveform. Typical unblanking signal waveforms are shown on figures 6 and 7.

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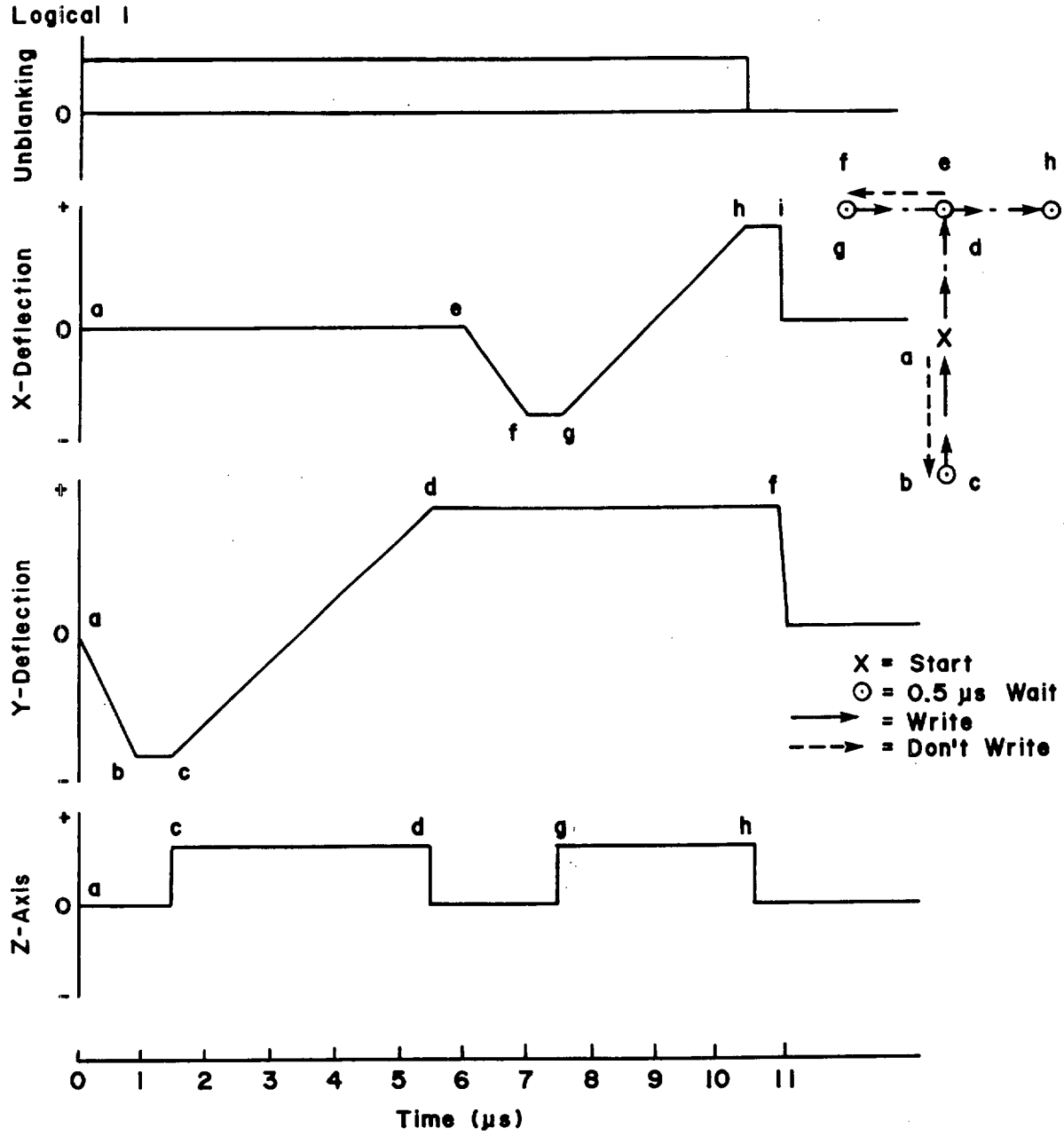


FIGURE 6. Typical deflection and video signal waveforms to form character "T".

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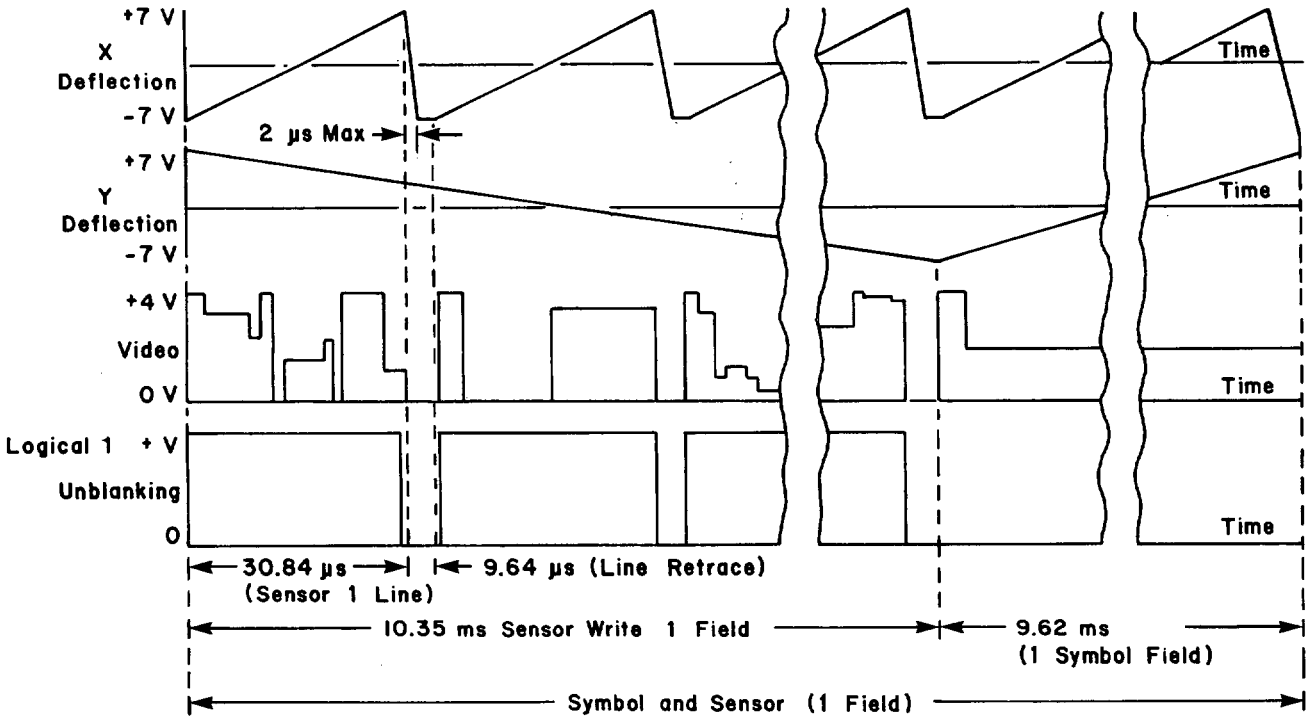


FIGURE 7. Typical deflection, video, and unblanking signal waveforms for symbol and raster modes.

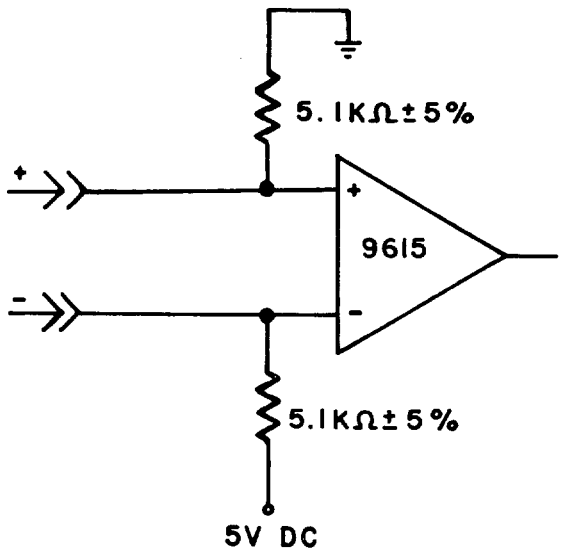


FIGURE 8. Converter-display differential receiver biasing for composite video, raster DO, self-test and unblanking inputs.

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3.5.1.11.4 Built-in-test (BIT) discrete. The BIT discrete signal provides an external request for self-test of the converter-display. The BIT discrete will be implemented with a balanced differential technique. Dedicated differential drivers will drive dedicated differential receivers via shielded twisted pair wires, (Logic "1" = Test; Logic "0" = Normal Operation). The driver and receiver characteristics shall be as specified in table II. Figure 8 shows the required differential receiver biasing.

TABLE II. Differential driver and receiver characteristics.

Signals	Characteristics
Line Driver	
Input High Voltage:	2.0 volts minimum
Input Low Voltage:	0.8 volt maximum
Input Down Current:	-1.6 mA maximum at 0.4 volt
Input Up Current:	100 uA maximum at 4.5 volts
Output High Voltage:	2.4 volts minimum at -15 mA
Output Low Voltage:	0.4 volt maximum at +15 mA
Line Receiver	
Input Differential Threshold:	±500 millivolts maximum
Input Common Mode Noise:	±15 volts maximum
Input Circuit Protection:	±20 volts line to ground or 20 volts differential line to line, without damage
Output High Voltage:	2.2 volts minimum at -5.0 mA
Output Low Voltage:	0.4 volt maximum at 15.0 mA
Input Current Low Level "TRUE" Input:	-0.7 milliampere maximum at 0.4 volt
Input Current High Level "TRUE" Input:	6.0 milliamperes maximum at 15 volts
Input Current Low Level "Complement" (Return) Input:	-1.6 milliamperes maximum at 0.4 volt
Input Current High Level "Complement" (Return) Input:	-7.9 milliamperes maximum at -15 volts

Logic 0 - "TRUE" side at low voltage level
"Complement" side at high voltage level
Less than 0.5 volt difference between "TRUE" and "Complement

Logic 1 - "TRUE" side at high voltage level
"Complement" side at low voltage level
"TRUE" exceeds "Complement" by greater than 0.5 volt positive

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3.5.1.11.4.1 Pulse width. The BIT discrete pulse is 128 us ±5 percent wide between its 90 percent points.

3.5.1.11.4.2 Pulse repetition rate. The converter-display shall be capable of receiving a BIT initiate signal once every refresh cycle.

3.5.1.11.5 Raster discrete. The raster discrete shall be used to command the converter-display to operate in a raster or symbol mode ("1" = raster, "0" = symbol). The raster discrete is a logical "1" or "0" as specified in table II. Figure 8 shows the required differential receiver biasing.

3.5.1.11.6 Composite video processing. The converter-display shall provide the capability to process an input composite video signal in response to a BACKUP RADAR MODE SELECT input discrete. Processing functions shall include sync-stripping to obtain sweep generator timing pulses, horizontal and vertical ramp generation, and video processing to provide the LAMPS Radar display. The resulting x, y, and z-axis signals required to generate the radar sensor image display shall be multiplexed into the deflection and video amplifier channels normally used to process input signals from the CMUX.

3.5.1.11.6.1 Signal characteristics. The composite video signal, including synchronizing and blanking pulses as shown on figures 9 and 10, shall be transmitted from the SDC to the converter-display over a triax cable. Signal parameters shall be as specified in table III or where additional detail is required the parameters shall be controlled by RS-343A.

3.5.1.11.6.2 Input impedance. The composite video signal channel input impedance shall be 93±3 percent ohms.

3.5.1.11.6.3 Converter-display input signal select default conditions. The converter-display input signal select circuitry shall default to accept only CMUX input signals under the following conditions:

a. Loss or absence of power in the signal data converter (SDC).

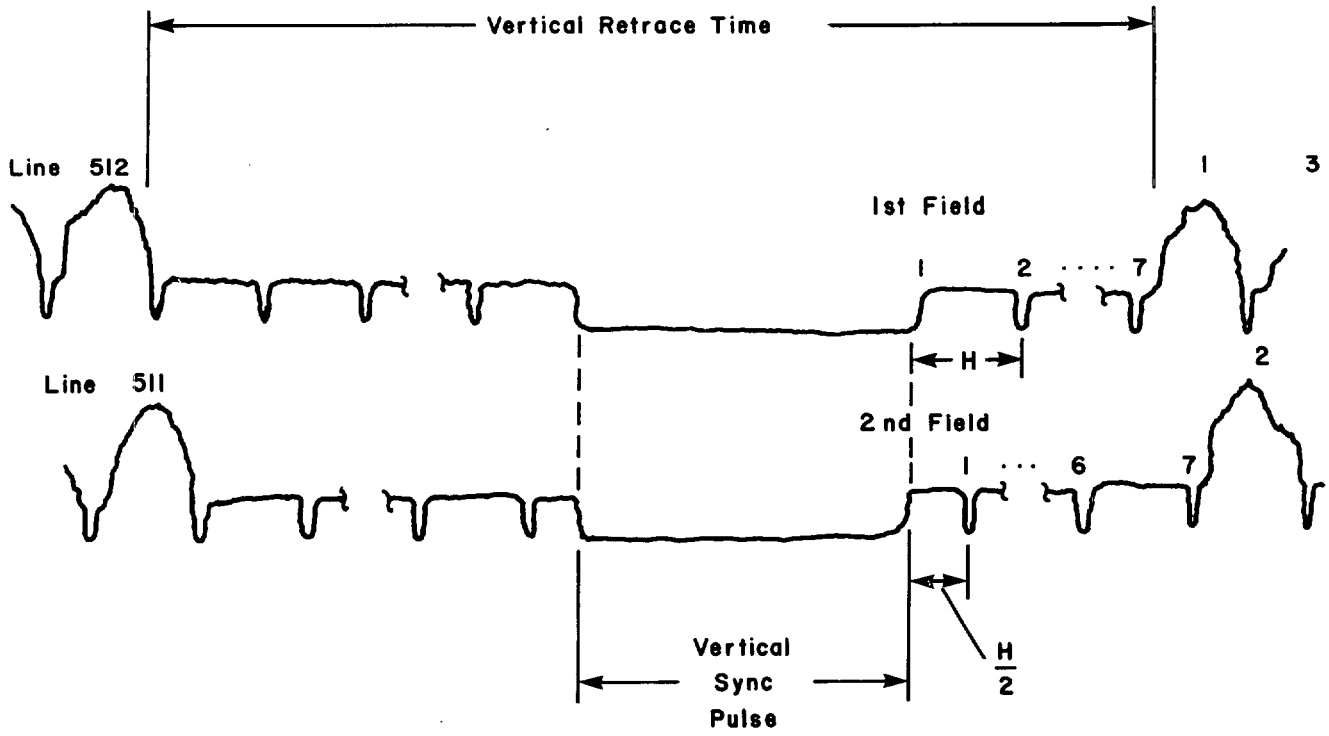
b. Absence or disconnect of the converter-display/SDC interconnecting cable.

c. Open circuit or loss of differential sustaining voltage on the backup radar mode select lines.

3.5.1.11.7 Backup radar mode select discrete. This discrete shall cause the converter-display to process the composite video input signal. In addition, this discrete shall be used as the RASTER DISCRETE, in accordance with 3.5.1.11.5, to command the converter-display to operate in the raster mode. The discrete shall also disable the BIT discrete input signal, specified in 3.5.1.11.4, from the CMUX. The converter-display shall be placed in a state of NORMAL OPERATION and periodic externally requested self-test will not be executed in the converter-display. No signal or other condition from the CMUX shall effect converter-display operation when in backup radar mode operation.

3.5.1.11.7.1 Signal characteristics. The backup radar mode select discrete shall be implemented with a balanced differential line in true/complement form sent to the converter-display from a 55113 line driver device in the SDC. The backup radar mode select shall be sustained by a logic "1" input

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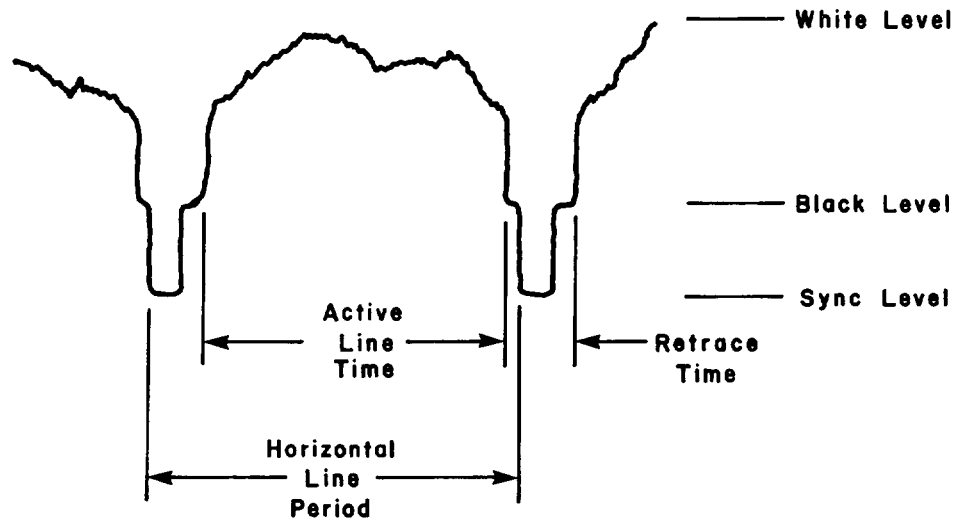


FIGURE 9. Composite video waveform.

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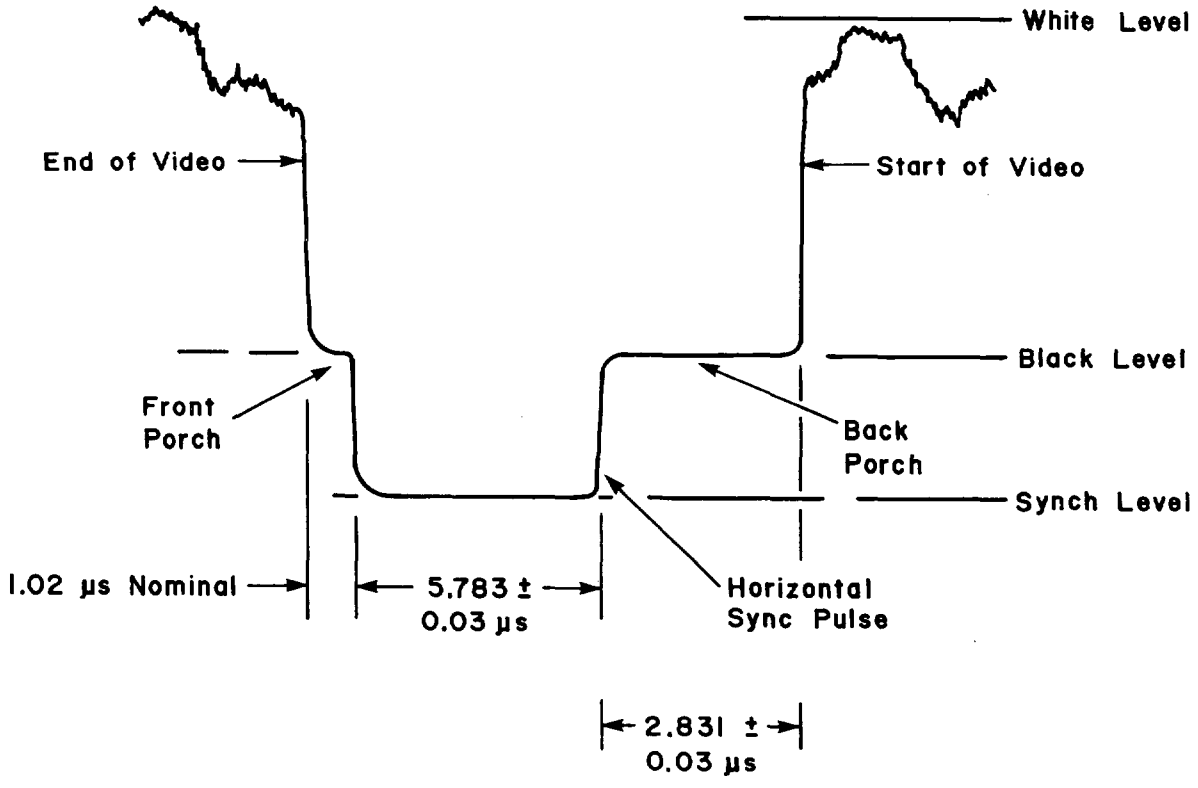


FIGURE 10. Horizontal sync waveform.

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TABLE III. Composite video signal parameters.

Signal	Parameter
Basic Timing	The basic timing shall be 493.57 horizontal line periods (H), per field (V) with two fields per frame.
Active Field Time	The active field time corresponding to 256 lines of video shall be 10.35±0.015 ms.
Vertical Retrace Time	The inactive field time corresponding to the vertical retrace period shall be 9.62±0.015 ms.
Vertical Period (V)	The time from the start of one field to the start of the next is the sum of the active and retrace times and shall be nominally 19.98±0.03 µs.
Horizontal Line Time	The active horizontal line times shall be 30.84±0.03 µs.
Horizontal Retrace Time	The horizontal retrace time shall be 9.64±0.03 µs.
Horizontal Period (H)	The time from the start of one line to the start of the next is the sum of the active and retrace line times and shall be 40.48±0.06 us.
Vertical Sync Pulse	The width of the vertical sync pulse shall be 131±5 us. The first line of video of the first field shall start 6 H periods after the trailing edge of the V sync pulse as shown on figure 8. The first line of video in the second field shall start 6-1/2 H periods after the V sync pulse trailing edge.
Horizontal Sync Waveform	The horizontal sync waveform shall consist of a front porch, a horizontal sync pulse and a back porch as shown on figure 9. The sum of the three components shall equal the line retrace time of 9.64±0.03 µs. The front porch shall have a width of 1.02 µs nominal. The horizontal sync pulse shall have a width of 5.783±0.03 µs. The back porch shall have a width of 2.831±0.03 µs.
Sync Stability	Vertical-to-vertical and horizontal-to-horizontal sync pulse period shall be stable to within 0.1 percent when measured on a pulse-to-pulse basis. The stability between the trailing edge of the vertical sync pulse and the horizontal sync pulse shall be ±60 nsec.
Line Stability	The period between the start of the 128th line of either field and the trailing edge of the vertical sync pulse shall be stable to within ±5 us.

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TABLE III. Composite video signal parameters - Continued.

Signal	Parameter
Voltage Levels	Reference levels shall have the following values: Black Reference Level = 0±0.1V White Reference Level = 4±0.1V relative to Black Level Sync Reference Level = -1.6±0.1V relative to Black Level

condition and logic "0" shall cause the converter-display to process all normal converter-display/CMUX interface signals shown on figure 2.

3.5.1.12 Performance characteristics. This section provides the performance characteristics of the video, deflection, blanking, and self-test circuitry required to generate the desired visual displays for symbol or raster modes.

3.5.1.12.1 Deflection channel performance. The performance characteristics of both the horizontal and vertical deflection channels are specified in 3.5.1.12.1.1 through 3.5.1.12.4.2.1. The performance characteristics of the converter-display horizontal and vertical deflection channels shall be required to provide raster scans on the CRT in accordance with the input sweep timing identified on figure 7 and the symbol deflections shown on figure 6.

3.5.1.12.1.1 Output signal amplitude. The output signal amplitude of the deflection channels shall be such that the electron beam is deflected 6.5 inches typically, given a 14V peak-to-peak input signal.

3.5.1.12.1.2 Deflection channel bandwidth. The deflection channel small signal bandwidth shall be not less than dc to 440 KHz at the 3 dB point with a pass-band flatness to within ±3 dB.

3.5.1.12.1.3 Noise modulation. Noise modulation of the CRT beam shall be less than 70 percent of the beam diameter at all times.

3.5.1.12.1.4 Gain adjust. Provision shall be made to adjust the deflection gain ±3 dB about the nominal value.

3.5.1.12.1.5 Deflection amplifier protection. The design of the deflection amplifier shall be such that signal voltages up to two full diameters deflection (28 Vp-p) shall not damage the deflection amplifier or the magnetic deflection yoke. Protection shall be provided when an input signal is present at the input to the deflection amplifier(s) which would cause damage to the amplifier.

3.5.1.12.1.6 CRT beam positioning accuracy. The relative error in geometrical CRT beam position with respect to the center of presentation shall be not greater than ±2 percent of the distance deflected in either axis. The relationship between input voltage and beam position shall be

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linear. The center of presentation shall be within ±.05 feet of the mechanical center of the CRT.

3.5.1.12.1.7 Beam positioning time. The CRT beam shall be positioned within 0.1 percent of the required position for the following beam displacements within the corresponding beam position delay time periods.

a.	Character-to-character	4 us
b.	Half screen	10 us
c.	Full diagonal	16 us

3.5.1.12.1.8 Deflection response to input sweep signals. Any overshoot, oscillation, distortion or delay of the electron beam deflection due to the initial excitation of the CRT by the linear input sweep signal shall stabilize to conform to the input driving signal within 0.75 us after initiation of the input sweep signal excluding deflection amplifier delay.

3.5.1.12.1.9 CRT raster horizontal retrace beam positioning time. The CRT beam shall be positioned within 0.5 percent of the next line start position within 8 us given an input horizontal retrace ramp fall time of not greater than 2 us.

3.5.1.12.2 Video channel performance requirements. The video channel performance requirements are specified in 3.5.1.12.2.1 and 3.5.1.12.2.2.

3.5.1.12.2.1 Dynamic range. The dynamic range of the video channel shall be adequate to produce eight ($\sqrt{2}$ type) shades of gray, including brightness full-on and brightness full-off under a diffused ambient illumination of 50 footcandles, and not less than four shades at 10,000 footcandles ambient. The input versus output function of the video channel shall be linear within ±10 percent using gamm correction as required measured over the input voltage range required to produce the eight shades of gray.

3.5.1.12.2.2 Video transient response. The overall rise and fall time response of the video channel shall be not greater than 16 us each when measured between the 10 percent and 90 percent points of the output pulse peak amplitude.

3.5.1.12.3 Unblanking channel performance. The unblanking channel shall exercise control of the video output.

3.5.1.12.3.1 Signal amplitude. The unblanking signal output level shall provide a voltage bias pedestal for the video signal at the CRT to preserve the capability of the external unblanking signal.

3.5.1.12.3.2 Unblanking channel transient response. The unblanking channel transient response shall be not greater than 100 ns between the 10 percent and 90 percent points.

3.5.1.12.3.3 Dynamic range. The unblanking signal provides the full dynamic range of the video channel during unblanking.

3.5.1.12.4 Test channel performance. The test channel shall be capable of two modes of operation. These two modes consist of manual self-test and

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external self-test. A photo diode CRT beam detection technique shall not be provided.

3.5.1.12.4.1 Manual self-test. During manual self-test, which is operator initiated by the test position of the "ON-OFF" switch on front of the converter-display, the equipment shall apply input voltages to deflection, video, and blanking channels in order to generate the test pattern shown on figure 11. During test pattern display, all external input signals shall be inhibited from being displayed. The test pattern shall provide the operator with a qualitative evaluation of converter-display performance as well as exercising the periodic BIT function.

3.5.1.12.4.2 External self-test. The automatic self-test shall be initiated by receipt of the BIT DO signal. Upon receipt of this signal, the converter-display shall commence tests of the internal electronic circuitry to verify proper operation of all channels. Upon completion of these tests, the test channel shall respond with a GO/NO-GO indication through the fault discrete. The performance requirements of the FAULT DI signal are provided in 3.5.1.14.1. The test channel shall respond to the automatic test request while the BIT discrete signal is in its "request" state. All internal tests must be completed and a response initiated within 128 us, ±5 percent after the BIT initated signal reaches its "request" state. The converter-display shall be capable of responding to the automatic test request immediately prior to each refresh cycle. If the converter-display experiences a NO-GO condition, the indicator shall be energized to the failure state and shall remain in this state until reset by a GO test result.

3.5.1.12.4.2.1 BIT initiate. Differential reception on twisted pair shall be provided with the characteristics identified in table II. True line definition "1" = Test Initiate and "0" = Normal Operation. This signal is a pulse 128 ±5 us wide which will be transmitted once every refresh period. The test shall be completed within the duration of this pulse.

3.5.1.13 Converter-display output requirements. The visual parameters of the converter-display and the test output signal shall be as specified in 3.5.1.13.1 through 3.5.1.13.10.

3.5.1.13.1 Brightness. The converter-display shall produce sufficient focused highlight brightness to produce a 3:1 contrast ratio in a diffused ambient illumination of 10,000 footcandles. The spectral quality of the ambient illumination shall approximate that of bright sky at midmorning or midafternoon. Contrast ratio is herein specified as CR = (B+D)/D where CR is the contrast ratio; B is brightness of the illuminated area in zero F.C. ambient; and D is the brightness of the nonilluminated area.

3.5.1.13.1.1 Brightness uniformity. The variation in display brightness shall be not greater than ±25 percent of the average highlight brightness in the symbol mode with input video variation that is proportional to input writing speed. The variation of brightness in the raster scan shall be not greater than 25 percent of the brightest portion of the raster in the absence of video modulation.

3.5.1.13.2 Contrast.

3.5.1.13.2.1 High ambient contrast. The contrast ratio of the display shall be not less than 3:1 where a small highlight area (bright line or dot) is compared to an unlighted dark area 0.2 inch away from the highlight spot

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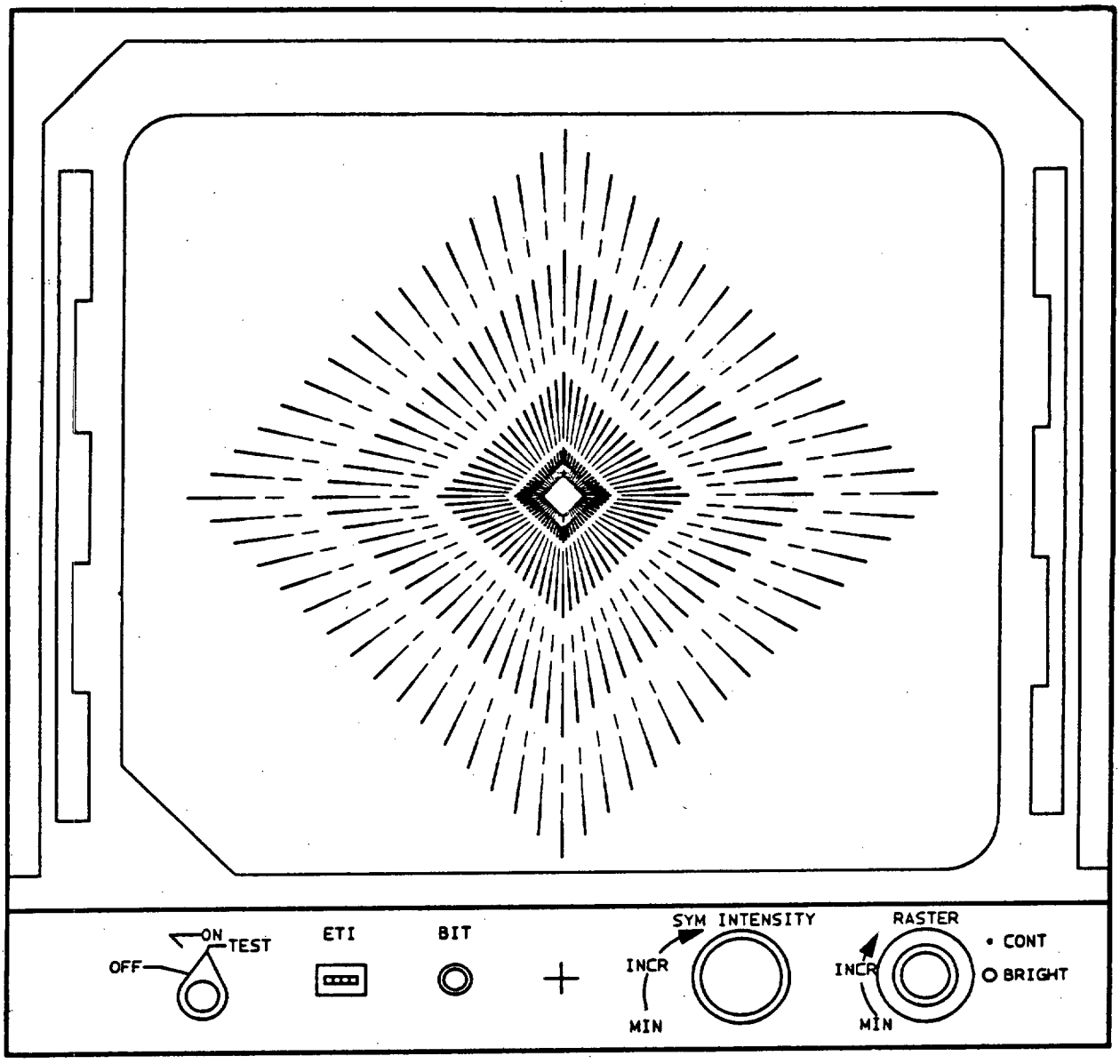


FIGURE 11. Manually initiated test pattern.

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and where the entire display face is illuminated by 10,000 footcandles of uniformly diffused ambient light.

3.5.1.13.2.2 Low ambient contrast. The contrast ratio of the display shall be not less than 12:1 where a small highlight area (bright line or dot) with an intensity equal to that intensity required to produce a contrast ratio of 3:1 in 10,000 footcandles ambient is compared to an unlighted dark area 0.2 inch away from the highlight spot and where the entire display face is illuminated with as close to zero footcandles as laboratory conditions allow.

3.5.1.13.3 Resolution. The width of a horizontal or vertical line at its half brightness points shall be not greater than 0.013 inch at the center of the tube face and shall not increase in width more than 30 percent of this dimension at any location away from the center. Measurement of resolution shall be made at a CRT beam writing speed of 210,000 inches per second with the intensity set at the brightness required for meeting the contrast ratio specified in 3.5.1.13.1. Focus control shall be adjusted for optimum focus. The display face shall be illuminated with 50 footcandles of uniformly diffused ambient light.

3.5.1.13.4 Jitter. The maximum motion of the CRT beam which is unrelated to the system deflection signals and which has a periodicity in the order of the frame rate shall be not greater than 0.007 inch.

3.5.1.13.5 Long term drift. After 5 hours of continuous operation, the drift in center position of the beam shall be not greater than 0.020 inch.

3.5.1.13.6 Flicker. There shall be no perceptible flicker when the converter-display is operated at 50 fields/second in a diffused ambient illumination of 50 to 10,000 footcandles. During raster scan, the converter-display shall be adjusted for 3:1 contrast as specified in 3.5.1.13.2.1 (high ambient) and 8:1 for low ambient.

3.5.1.13.7 Phosphor. The type of phosphor shall be selected to meet the performance requirements specified herein. Type P-43 or other phosphor with superior characteristics may be recommended, with justification, for acceptance by the procuring activity.

3.5.1.13.8 Phosphor protection. The converter-display shall contain circuitry which will cause the CRT beam to be cut off with any combination of manual or automatic control settings for all conditions which could result in phosphor burning. The protection circuit shall not turn off the beam under the normal conditions for displaying a 0.125 inch or larger radius circle, or any normal input deflection signal.

3.5.1.13.9 Surface reflection. The CRT face shall be provided with a reflection reducing coating conforming to the requirements of MIL-C-14806 in addition to withstanding the environmental conditions specified herein.

3.5.1.13.10 Deflection non-linearity. The deflection non-linearity shall be not greater than $\pm$2 percent of the length of any line at any point within the central 80 percent of viewing area and shall be not greater than $\pm$5 percent in the remaining area.

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3.5.1.14 Output signals.

3.5.1.14.1 Fault discrete signal. The fault discrete signal shall be sent from a differential transmitter in the converter-display to a differential receiver in the CMUX. A logic "1" designates FAULT and a logic "0" designates NORMAL OPERATION.

3.5.1.14.2 Overtemperature signal. The overtemperature signal and return specified in 3.3.13.1, shall be connected to two pins on the J2 connector as a contact closure.

3.5.1.15 Converter-display performance requirements summary. Table IV shows a summary listing of the converter-display performance requirements identified in 3.5.1 and 3.5.2, as well as other performance characteristics which are essential to meet the requirements of the LAMPS MK III program.

TABLE IV. Converter-display requirements summary.

Characteristic	Ref. Para.	Parameter Requirements
CRT	3.5.1.5.2 3.5.1.5.6	Electromagnetic shield implosion protected
Surface Reflection Reduction	3.5.1.13.9	Anti-reflective coating
Viewable Display Area	3.5.1.8	6.5" x 6.5"
Contrast	3.5.1.13.2	3:1 with 10,000 footcandles ambient; 12:1 with 50 footcandles ambient
Area Brightness Uniformity	3.5.1.13.1.1	25% variation maximum, for symbols and rasters
Phosphor	3.5.1.13.7	P43 or superior
Phosphor Protection	3.5.1.13.8	As required
Refresh (Flicker)	3.5.1.13.6	50 Hz
Line Width	3.5.1.13.3	0.013 inch at center: +30% elsewhere
Gamma Correction	3.5.1.12.2.1	As required to provide ±10% video channel linearity for 8 shades
Beam Jitter	3.5.1.13.4	0.007 inch maximum
Relative Positioning Accuracy	3.5.1.12.1.6	±2% any direction w.r.t. center of presentation
Positional Stability	3.5.1.13.5	0.02 inch drift (maximum) over 5 hours

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TABLE IV. Converter-display requirements summary - Continued.

Characteristic	Ref. Para.	Parameter Requirements
Deflection Factor Non-Linearity	3.5.1.13.10	±2% in central 80% of display area and 5% elsewhere
Video Bandwidth	3.5.1.11.2.2	DC to 27 MHz (3 dB)
HV Protection	3.5.1.5.7	Arc suppressors
Warm-up	3.3.11	2 minutes (maximum) (standard condition)
Flicker	3.5.1.13.6	Not perceptible at 50 fields/Sec in 50 to 10,000 footcandles ambient illumination
CRT Beam Writing Speed	3.5.1.13.3	210,000 inches/second Maximum (at 3:1 contrast and 10,000 footcandles ambient)
Deflection Scale Factor	3.5.1.12.1.1	7 volts = 3.25 inches
MTBF	3.3.2.4	2600 hours
Periodic Built-In-Test	3.3.7.6.1	90% of all failures detected converter-display alone; 98% detection with CMUX
Gray Shades ($\sqrt{2}$)	3.5.1.12.2.1	4 at 10,000 footcandles ambient; 8 at 50 footcandles ambient
Noise Modulation of CRT Beam	3.5.1.12.1.3	70% of beam diameter maximum
CRT Random Beam Positioning Time to 0.1%	3.5.1.12.1.7	4 us character-to-character in type mode, 10 us per half screen, 16 us full diagonal
CRT Raster Horizontal Retrace Beam Positioning	3.5.1.12.1.9	8 us maximum, (Assumes Input ramp fall time of 2 us maximum) time to 0.5%
Composite Video Backup Mode	3.5.1.11.6	Modified 975 line raster

3.5.2 Converter-display mounting tray. The converter-display mounting tray shall conform to the requirements specified in 3.5.2.1 through 3.5.2.3.

3.5.2.1 Function (mounting tray). The converter-display mounting tray shall permit the converter-display to slide in and out of the mounting tray,

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permit the converter-display carrying handle to jack the converter-display in and out of the mounting tray and connector, and permit locking the converter-display in place.

3.5.2.2 Form factor (mounting tray). The converter-display tray form factor and other dimensions shall be in accordance with Drawing 2845951 and as shown on figure 12. The size of the tray excluding connector shall be not greater than the following dimensions:

a. Width 9.24 inches

b. Height 2.25 inches

c. Depth 15.46 inches

3.5.2.3 Weight. The total weight of the converter-display tray excluding cables, shall be not greater than 1.8 pounds.

3.6 Workmanship. Workmanship of the converter-display shall conform to MIL-STD-454, Requirement 9, and the following:

a. Printed wiring board coatings shall be free of dirt and metal particles.

b. Chemical coated and painted surfaces shall have uniformly coated surfaces.

c. Metal parts shall be free of burrs or sharp edges.

d. The final assembly shall be free of dirt, solder fragments, hardware, or other foreign objects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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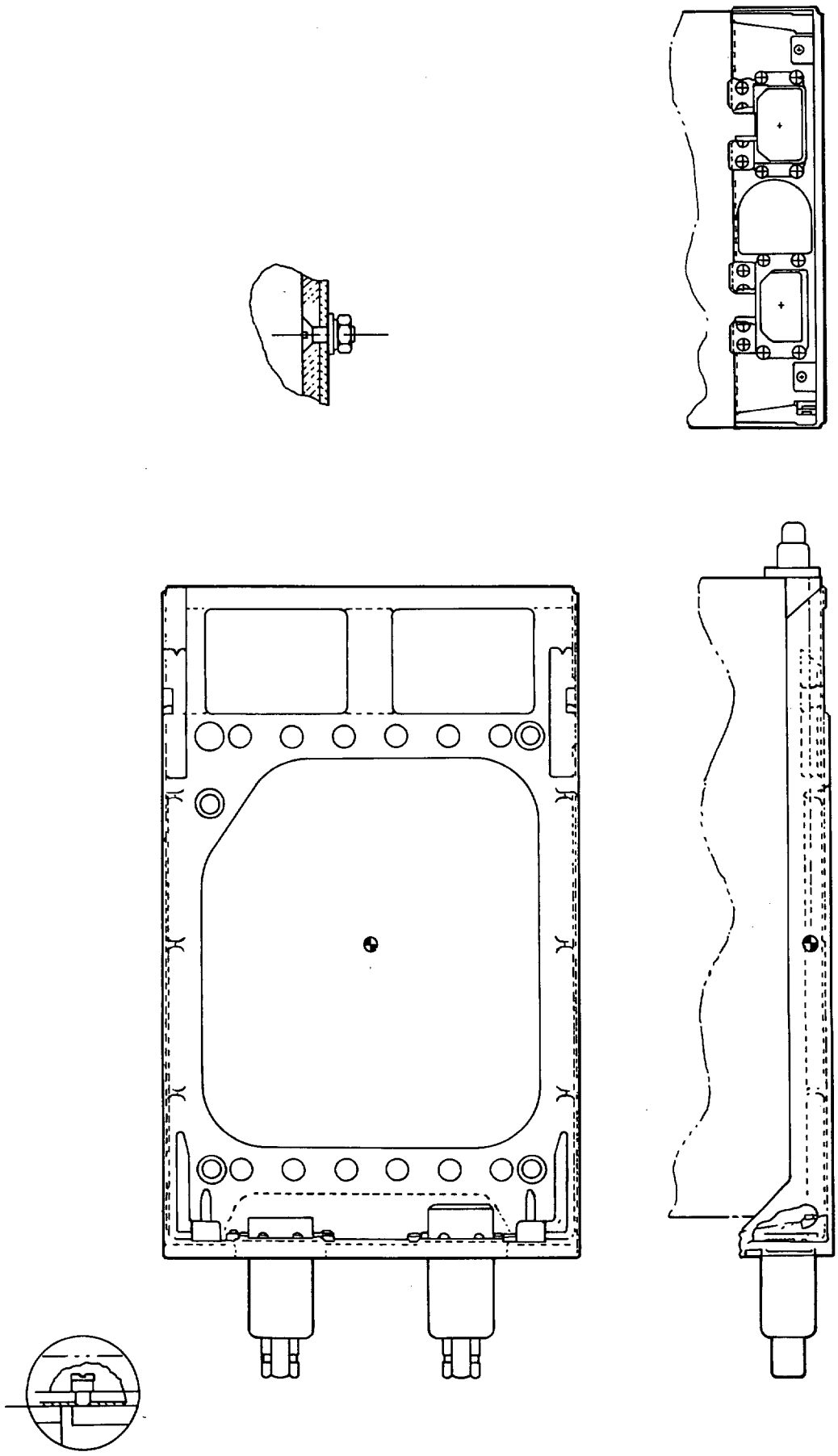


FIGURE 12. Outline/installation drawing, mounting tray, converter-display.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.4).

b. Initial production tests (see 4.5).

c. Quality conformance inspections (see 4.6).

4.3 Inspection conditions. Unless otherwise specified in the contract, all inspections shall be performed in accordance with the test conditions specified in 3.3.9 and 3.3.10.

4.4 First article inspection. First article inspection shall be conducted on converter-displays representative of the production control-indicator sets to be supplied under the contract. First article test performance shall be the responsibility of the contractor and shall be performed in accordance with approved test procedures. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired.

4.4.1 Scope of first article tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the converter-displays meets all the requirements of this specification and the contract. First article tests shall provide evidence of compliance with each requirement of Section 3. Table V is provided as a guide for cross-reference. First article tests shall include, but not necessarily be limited to the following:

TABLE V. Inspection requirements.

Requirements	I. First Article Inspection	II. Quality Conformance Inspections		
		Individual (A)	Sampling (B)	Special (C)
Visual Examination (3.2, 3.3.1, 3.5.1.2)	4.6.1.1	4.6.1.1	4.6.1.1	4.6.1.1
Performance Characteristics (3.4)	4.4.2	4.6.1.2	4.6.1.2	4.6.1.2
Workmanship (3.6)	4.6.2.2.1	4.6.2.2.1	4.6.2.2.1	4.6.2.2.1
Reliability (3.3.2)	4.6.2.1.1	4.6.2.1.2		
Temperature (3.3.10.1.1, 3.3.10.1.2)	4.4.3.1.1, 4.4.3.1.2	1/	2/	3/

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TABLE V. Inspection requirements-Continued.

Requirements	I. First Article Inspection	II. Quality conformance Inspection		
		Individual (A)	Sampling (B)	Special (C)
Temperature Shock (3.3.10.1.3)	4.4.3.8			
Low Pressure (3.3.10.2)	4.4.3.2			
Explosion Proof (3.3.10.3)	4.4.3.7			
Salt Fog (3.3.10.4)	4.4.3.6			
Random Vibration (3.3.10.5)	4.4.3.3			
Shock (3.3.10.6.1, 3.3.10.6.2, 3.3.10.6.3)	4.4.3.4	1/	2/	3/
Humidity (3.3.10.7)	4.4.3.5			
Warm-up Time (3.3.11)	4.4.4			
Electromagnetic Interference (3.3.6)	4.4.5			

1/ Additional individual tests may be requested by the procuring activity (see 4.6.1).

2/ Sampling tests to be performed will be determined by the procuring activity base on first article and individual test results (see 4.6.3).

3/ Special tests to be performed will be determined by the procuring activity (see 4.6.4).

a. Electrical (bench) (see 4.4.2).

b. Environmental (see 4.4.3).

c. Electromagnetic interference (see 4.4.5).

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d. Reliability development (see 4.4.6).

e. Maintainability demonstration (see 4.4.7).

4.4.2 Electrical (bench) tests. Bench tests shall be conducted to demonstrate compliance with specified performance parameters. These tests shall be conducted under normal laboratory conditions. Inspection and examination to determine compliance with material and workmanship requirements shall be conducted at this time. High and low temperatures shall be used to evaluate temperature dependent parameters.

4.4.3 Environmental tests. Environmental tests shall be conducted in the sequence specified in MIL-T-5422. Environmental tests shall include, but not be limited to, 4.4.3.1 through 4.4.3.8.

4.4.3.1 Temperature.

4.4.3.1.1 High and low temperature (operating). The converter-display shall be subjected to high and low temperature tests in accordance with MIL-STD-810, Method 501.2, Procedure I, and MIL-STD-810, Method 502.2, Procedure II, respectively, with the following modifications:

a. Continuous operation:

(1) High temperature extreme +71°C.

(2) Low temperature extreme -40°C.

b. Intermittent operation:

(1) High temperature extreme +95°C (edgelit panel lights shall not be operated at temperatures above +71°C ambient.

(2) Low temperature extreme -40°C.

4.4.3.1.2 High and low temperature (storage). The converter-display shall be subjected to high and low temperature storage tests in accordance with MIL-STD-810, Method 501.2, Procedure I, and MIL-STD-810, Method 502.2, Procedure I, with the following exceptions. High temperature extreme shall be +85°C, and low temperature extreme shall be -62°C.

4.4.3.2 Low pressure (altitude). The converter-display shall be subjected to low pressure (altitude) test in accordance with MIL-STD-810, Method 500.2, Procedure II. Altitude extreme shall be 40,000 feet.

4.4.3.3 Vibration test. A vibration test shall be performed with the converter-display and mounting tray hard mounted to a fixture using screws in lieu of the normal means of attachment as specified in Procedure I, Part 1 of MIL-T-5422, and in accordance with Curve IIIb of MIL-E-5400 except that the lower test frequency shall extend to 4 Hz and between 4 Hz and 7 Hz the acceleration shall be 0.5g. During resonant survey it shall be verified that no resonant frequencies exist below 50 Hz, as specified in 3.3.10.5. In addition to the test procedure of MIL-T-5422, it is required that during resonance survey the following vibration frequency bands must be closely monitored to assure that resonance at these frequencies does not result in converter-display failure or reduced reliability. Unless otherwise specified in the contract, frequencies to be used for endurance tests shall

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consider both converter-display resonance frequencies and the following critical frequency ranges (see 6.2.1):

a. 4 to 5 Hz.

b. 16 to 23 Hz.

c. 32 to 40 Hz.

d. 48 to 60 Hz.

e. 72 to 80 Hz.

f. 120 to 130 Hz.

g. 328 to 338 Hz.

4.4.3.4 Shock test. Shock test shall be performed without vibration isolators (referred to as direct hard mounting) and in accordance with MIL-T-5422 and shall include the following tests:

a. Basic Design Test, Part I, Figure 10.

b. Crash Safety Test, Part II, Figure 10.

c. Bench Handling Test.

4.4.3.5 Humidity test. A humidity test shall be performed in accordance with MIL-T-5422.

4.4.3.6 Salt fog test. A salt fog test shall be performed in accordance with MIL-T-5422.

4.4.3.7 Explosion proof test. An explosion proof test shall be performed in accordance with Procedure I of MIL-T-5422, except that the upper operational altitude limit shall be 15,000 feet.

4.4.3.8 Temperature shock test. A temperature shock test shall be performed in accordance with MIL-T-5422.

4.4.4 Warm-up time. The converter-display shall be subjected to tests to determine the capability of the converter-display to conform to the warm-up time requirements of 3.3.11.

4.4.5 Electromagnetic compatibility test. Equipments shall be tested in accordance with the requirements of MIL-STD-461 as modified by 3.3.6 and in accordance with the measurement procedures of MIL-STD-462.

4.4.6 Reliability development test. A reliability development test shall be conducted in accordance with MIL-STD-2068 with the following modifications:

a. The cumulative operating time shall be a minimum of 5,000 unit hours or 11 times the specified in MTBF, whichever is less.

b. No sample unit shall be operated for less than one-half the average operating time of all sample units on test.

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c. Accept Criteria. The converter-display shall be judged to have completed the RDT as required when the total operating time has been completed. Acceptance shall be upon procuring activity approval of corrective action on all failures occurring during the test.

A minimum of two converter-displays shall be tested. Test conditions shall be in accordance with Test Level F of MIL-STD-781 except the lower temperature limit shall be -40°C instead of -54°C and vibration shall be applied at least 30 minutes during each hour of converter-display operation.

4.4.7 Maintainability demonstration tests. A maintainability demonstration shall be conducted in accordance with MIL-STD-471, Method 8. Maintainability indices shall be verified in accordance with MIL-STD-2084. In addition to requirements of MIL-STD-2084, the demonstration of Non-Ambiguity Ratio, Shop Non-Ambiguity Ratio, Fixed Interface Ratio, and Shop Fixed Interface Ratio shall be included in the maintainability demonstration conducted in accordance with MIL-STD-471, Method 8. The Non-Ambiguity and Shop Non-Ambiguity Ratio shall be demonstrated by verifying that the BIT and test points are adequate to detect and isolate induced faults to meet specification requirements. A separate demonstration shall be performed for organizational and intermediate maintenance.

4.4.7.1 Selection faults. Sample size and fault selection shall be in accordance with MIL-STD-471. Built-In-Test Equipment (BITE) failures shall be considered as candidates for fault selection. The procuring activity will select a random sample fault for each demonstration.

4.4.7.2 Accept/reject criteria. The accept/reject criteria for the demonstration of corrective maintenance times shall be in accordance with MIL-STD-471, Method 8. An accept decision can only be made when an accept decision is made for all indices.

4.4.7.3 Acceptance. The maintainability demonstration shall demonstrate that the converter-display has met the accept criteria prior to the delivery of any converter-display.

4.4.7.4 Rejection. If a reject decision is reached, the procuring activity concerned shall be immediately notified. The contractor shall take the following action:

a. Develop a plan for design or material changes adequate to correct all deficiencies.

b. Upon approval of the plan, perform corrective action and repeat the demonstration until an accept decision is reached.

4.4.8 First article approval. Approval of the first article sample shall be by the procuring activity upon completion of all tests as required. No production converter-displays shall be delivered prior to the approval of the first article sample.

4.4.9 Equipment failure (first article). Should a failure occur during first article testing, the following action shall be taken:

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a. Notify the Government Inspector immediately.

b. Determine the cause of failure.

c. Implement and verify corrective action and continue first article tests.

The procuring activity reserves the right to repeat any tests necessary to verify the corrective action.

4.5 Initial production tests. One of the first ten production converter-displays shall be selected for tests by the procuring activity after the converter-display has successfully passed all individual tests. No other tests shall be conducted on converter-display prior to starting the initial production tests. A first article sample shall not be selected for this test.

4.5.1 Scope of initial production tests. This converter-display may be subjected to any and all tests the procuring activity deems necessary to assure that the production converter-display is equivalent to the previously approved first article sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.5.2 Accessory material. In addition to the complete converter-display submitted for initial production tests, the contractor shall also submit accessory material and data necessary to test the converter-display.

4.5.3 Initial production sample approval. Approval of the initial production sample shall be by the procuring activity upon completion of all required tests.

4.5.4 Reconditioning of initial production test sample. On completion of the initial production test, the converter-display shall be reworked by the contractor by replacing all limited life or damaged items. After reworking, the contractor shall resubmit the converter-display for acceptance.

4.6 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the Government inspector. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following (see table V):

a. Individual tests (see 4.6.1 and table V).

b. Sampling tests (see 4.6.2 and table V).

c. Special tests (see 4.6.3 and table V).

4.6.1 Individual tests. Each converter-display submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each converter-display accepted shall have passed the following tests:

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a. Examination of product (see 4.6.1.1).

b. Operational test (see 4.6.1.2).

c. Production reliability acceptance test (see 4.6.2.2).

4.6.1.1 Examination of product. Each converter-display shall be examined carefully to determine that the material and workmanship requirements have been met. Attention shall be given, but not limited to the following:

a. Completeness.

b. Nameplates, identification markings, and labels.

c. Finishes.

d. Welded and soldered joints.

e. Fit of components.

f. Mounting of components.

g. Safety features and interlocks.

h. Fastening and securing of devices and parts.

i. Cables, plugs, and receptacles.

j. Grounding connections.

k. Weight.

l. Overall dimensions.

m. Potential shorts and defective insulation.

n. Foreign objects, lint, dust.

o. Physical characteristics of the converter-display agree with the outline drawings specified.

4.6.1.2 Operational test. Each converter-display shall be operated not less than 5 minutes to permit the temperature to stabilize and to check sufficient characteristics and record adequate data to assure specified equipment operation.

4.6.2 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (sampling) phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and MIL-STD-2084.

4.6.2.1 Qualification phase. The qualification phase for the first article and production converter-displays shall be as specified in 4.6.2.1.1 through 4.6.2.2.1.1.

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4.6.2.1.1 Preproduction. Prior to the acceptance of converter-displays under the contract or order, the converter-display shall be tested as specified in MIL-STD-781, Test Plan IV, Test Level F (except that the lower temperature limit shall be -40°C instead of -54°C) and vibration shall be applied not less than 30 minutes during each hour of operation.

4.6.2.1.2 Production. Prior to the acceptance of converter-displays under the contract or order, a minimum of three converter-displays shall be tested as specified in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of converter-displays to be used shall be those listed in Table 5 of MIL-STD-781. For the qualification phase, Test Level F shall be used, except that the lower temperature limit shall be -40°C instead of -54°C and vibration shall be applied not less than 30 minutes each hour of operation. The accept/reject criteria for Test Plan III shall be used.

4.6.2.2 Reliability production acceptance phase. The converter-displays, spares, and repair parts throughout first article and production, shall be tested as specified in MIL-STD-781 (as modified herein) under Section 4.2.9, entitled "Test Plan XXIX (All Equipments Screening Test)". Classification of failures shall be in accordance with MIL-STD-781 and MIL-STD-2074. The reject criteria of Test Plan XXIX shall not apply to first article converter-displays.

4.6.2.2.1 All equipment test. Each converter-display, spare, and repair part produced except those submitted for the Production Reliability Qualification and the Preproduction Reliability Development Tests shall be tested under the requirements of Test Level F of MIL-STD-781 (except the lower temperature limit shall be -40°C instead of -54°C) for a period not less than 100 hours failure-free. A failure shall be specified as any out-of-specification performance or any condition which requires repair or adjustment of the converter-display under test. Only adjustment of knob-type controls that are readily accessible to the operator during normal use will be permitted. Prior to the 100 hour test on each converter-display, a random vibration stress screening shall be performed on each production unit. The converter-display shall be hard mounted (without shock isolators) and subjected to an overall level of 6.38g RMS ±10 percent for an accumulated time of 10 minutes in the plane perpendicular to the majority of the converter-display printed wiring assemblies. The vibration frequency range and power spectral density shall be as follows:

a. 20 - 200 Hz +3 dB/Octave

b. 200 - 700 Hz $0.04g^2/Hz$

c. 700 - 2000 Hz -6.6 dB/Octave

During vibration, the unit shall be energized and operated with its inputs stimulated and outputs monitored. All failures occurring during the random vibration stress screening shall be corrected and the vibration resumed to accumulate the 10 minutes. Vibration stress screening test time and failures shall not be used in determining MTBF compliance. Prior to the 100 hour test on each converter-display, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used, the details thereof must be included in the approved test procedure. During the 100 hour period of failure-free operation the converter-display shall be exercised and tested in its various modes of operation. Should a failure occur, the converter-display

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shall be repaired, proper operation verified, and the test continued. The accumulation of failure-free time shall start at zero time upon resumption of testing. Corrective action shall be taken should failures occur, as specified in MIL-STD-7081. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of converter-display at any time that a reject situation exists pending a review of the contractor's efforts to improve the converter-display, the converter-display parts, or the converter-display workmanship so that the entire compilation will show other than reject decision.

4.6.2.2.1.1 Safety test/inspection. A ground integrity check shall be performed on the Primary Power Circuit (PPC) to ensure a maximum of 0.1 ohm exists between all PPC grounds and the chassis. This is based on using a digital multimeter (Fluke 8000A or equivalent/better). An isolation check shall be performed on the PPC to ensure a minimum of 10 megohms exists between each input power line and the PPC ground. This is based on using a digital multimeter (Fluke 8000A or equivalent/better). A safety inspection shall be performed to ensure all PPC grounds are correctly assembled and mechanically stable. All PPC wiring and components shall be inspected for damage. A Hi-Pot test shall be performed on the PPC and must be capable of withstanding without breakdown a potential of 500 Vdc for a duration of 60 seconds. The test potential may be gradually increased to the required value, but the full value is to be applied for 60 seconds. The test voltage must be applied with all current carrying input power lines connected (tied together) to the positive side of the Hi-Pot tester and the negative side connected to ground (dead metal parts). The leakage current shall be not greater than 5 ma. An inspection for damage shall be performed following the conclusion of the test. Any rework in the PPC after Hi-Pot testing shall require an additional performance of the operations listed above before delivery. This shall also include field failures which have rework performed on the PPC.

4.6.3 Sampling tests. Converter-displays selected for sampling tests shall first have passed the individual tests. (Sampling tests shall not be conducted unless reliability assurance tests of 4.6.2 are deleted by contractual action.) Converter-displays will be selected for sampling tests by the Government inspector in accordance with table VI.

TABLE VI. Sampling tests sample selection.

Quantity of equipments offered for acceptance	Quantity to be selected for sampling tests
1-50	1
51-125	2
126-225	3
226 and above	3 1 for each additional 150 or fraction thereof.

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4.6.3.1 Scope of sampling tests. As a minimum, each converter-display selected for sampling tests shall be subjected to the following:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each converter-display tested and should be based on results of the first article, individual, and special tests.

4.6.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the converter-display and to assure quality control.

4.6.4.1 Special test schedule. Selection of converter-displays for special tests shall be made as follows:

a. On early converter-displays after an engineering or material change.

b. Whenever failure reports or other evidences indicate that additional tests are required. (This will be determined by the procuring activity.)

4.6.4.2 Scope of special tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure acceptable to the procuring activity prior to conducting the tests.

4.6.5 Equipment failure (special tests). Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

a. Notify the Government Inspector immediately.

b. Determine the effectivity of BIT to detect and isolate the failure.

c. Determine the cause of failure.

d. Determine if the failure is design defect.

e. Implement a corrective action to eliminate recurrence of failure.

f. Where practical, include a test (in the individual test) to check all converter-displays for this requirement until assurance is obtained that the defect has been corrected.

4.7 Production equipments. Converter-displays supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved first article sample. Each converter-display shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection, and for equipment already accepted by the Government. It shall be the obligation of

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the contractor to make necessary corrections as approved by the procuring activity.

4.8 Reconditioning of tested equipment. Unless otherwise specified in the contract, converter-displays which have been subjected to first article, initial production, reliability and special tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the converter-displays for acceptance (see 6.2.1).

4.9 Presubmission testing. No item, part, or complete converter-display shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all requirements.

4.10 Rejection and retest. Unless otherwise specified in the contract, converter-displays which have been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance (see 6.2.1). Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

4.11 Methods of examination and test. The procuring activity or Government inspector reserves the right to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.2).

4.12 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of Section 5 and the standards and specifications listed therein.

5. PACKAGING

5.1 General. All major units and parts of the converter-display shall be preserved, packaged, packed and marked for the level of shipment specified in the contract, or order, in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The CV-3252/A, converter-display specified herein will form a part of the LAMPS MK III Ship/Air System which is to be mounted and operated within a helicopter, working off a destroyer type ship.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. When a first article is required (see 4.4)

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c. If the critical frequencies specified in 4.4.3.3 are to be used in vibration endurance tests.

d. If converter-displays subjected to tests (or converter-displays rejected) may be reworked and resubmitted for acceptance (see 4.8 and 4.10).

e. Selection of applicable levels of packaging and packing (see section 5)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirements title	Applicable DID no.	Options
4.4.1	First Article Inspection Report	DI-T-4902	---
4.4.9, 4.6.5	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719	---
4.4.9, 4.6.5	Report, Failure	DI-R-21598	---
4.11	Production/Acceptance Inspection Procedures	DI-T-4903	---
4.11	First Article Inspection Procedures	DI-T-4901	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the converter-display should be a first article sample. The first article should consist of two converter-displays. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Associated equipment. The converter-display specified herein should be designed to operate with the AN/AYK-14 Digital Data Computer.

6.5 Subject term (key word) listing.

Converter-Display CV-3252/A
Destroyer type ship
Helicopter
LAMPS MK III Ship/Air System
Raster display mode

Preparing Activity:
Navy - AS
(Project 5836-N001)

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APPENDIX A

INTERFACE WIRING REQUIREMENTS
FOR
CONVERTER-DISPLAY CV-3252/A

10. SCOPE

10.1 Scope. This appendix establishes the interface wiring requirements for the CV-3252/A Converter-Display. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Connector pin designations. The connector pin designations for the converter-display are specified in tables VII, VIII, and IX.

TABLE VII. Test signal connector pin designations.

Connector No.: J-1		Connector Type: MS27656T19B35S
Pin	Function	Source
1	PRDY	A1-J4-13
2	HVPSEN	A1-J4-18
3	KOUT	A1-J4-16
4	G1OUT	A1-J4-17
5	XFB	A1-J4-36
6	YFB	A1-J4-35
7	FILSNS	A1-J4-34
8	TSTEN	A1-J4-37
9	ANSNS	A1-J4-07
10	FOCSNS	A1-J4-26
11	G2SNS	A1-J4-25
12	+80V	PS2-J2-FF
13	-80V	PS2-J2-A
14	+22V	PS2-J2-W
15	-22V	PS2-J2-C
16	+65V	PS2-J2-Y
17	+15V	PS2-J2-P
18	-15V	PS2-J2-N
19	+5V	PS2-J2-Z
20	6.32V	PS2-J2-F
21	RTN	PS2-J2-BB

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TABLE VIII. JS-2 signal connector pin designations.

Connector No.: J-2		Connector Type: DPXAME-32W4-33S-FO
Pin	Function	Source
32	X DEFLECTION	CMUX
31	Y DEFLECTION	CMUX
30	VIDEO	CMUX
6	CD FAULT DI RETURN	Converter-Display
7	CD FAULT DI	Converter-Display
2	CD OVERTEMP DI RETURN	Ground
1	CD OVERTEMP DI	Converter-Display
13	SELF-TEST DO	CMUX
14	CD RASTER DO	CMUX
15	UNBLANKING SIGNAL	CMUX
20	SELF-TEST RTN	CMUX
21	CD RASTER DO RETURN	CMUX
22	UNBLANKING SIGNAL RETURN	CMUX
27	SHIELDS OF TWISTED PAIRS	CMUX
28	OUTER TRIAX SHIELD	CMUX
29	COMPOSITE VIDEO	Signal Data Converter
17	BACKUP RADAR MODE SELECT	Signal Data Converter
18	BACKUP RADAR MODE SELECT RETURN	Signal Data Converter

TABLE IX. JS-3 power connector pin designations.

Connector No.: J-3		Connector Type: DPXAME-32W4-34P-FO
Pin	Function	Source
1	115 Vac ØA 400 Hz	Helicopter
2	115 Vac ØB 400 Hz	Helicopter
10	115 Vac ØC 400 Hz	Helicopter
5	Neutral (Airframe Ground)	Helicopter
7	0 to 5 Vac	Panel Lamp Control
6	0 to 5 Vac Return	Panel Lamp Control
13	Safety Ground	Helicopter
8	Chassis Safety Ground	Helicopter
9	Chassis Safety Ground	Helicopter
23	Chassis Safety Ground	Helicopter
24	Chassis Safety Ground	Helicopter
3	115 Vac 400 Hz Phase A	Converter-Display
4	115 Vac 400 Hz Phase B	Converter-Display
12	115 Vac 400 Hz Phase C	Converter-Display

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APPENDIX B

EXCEPTIONS TO MIL-STD-1472
APPLICATION TO
CONVERTER-DISPLAY CV-3252/A

10. SCOPE

10.1 Scope. This appendix specifies the exceptions to MIL-STD-1472 applicable to the Converter-Display, CV-3252/A. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. REQUIREMENTS

30.1 Design exceptions. Table X specifies design exceptions to MIL-STD-1472 applicable to the converter-display.

TABLE X. Exceptions to design requirements of MIL-STD-1472.

Specification	Paragraph	Exception
MIL-STD-1472	3.3.15	Decrease the knob spacing requirement from 1.0 inch (minimum) to .50 inch (minimum) between the SymInten and Raster Bright controls.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85872(AS)	CONVERTER-DISPLAY CV-3252/A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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